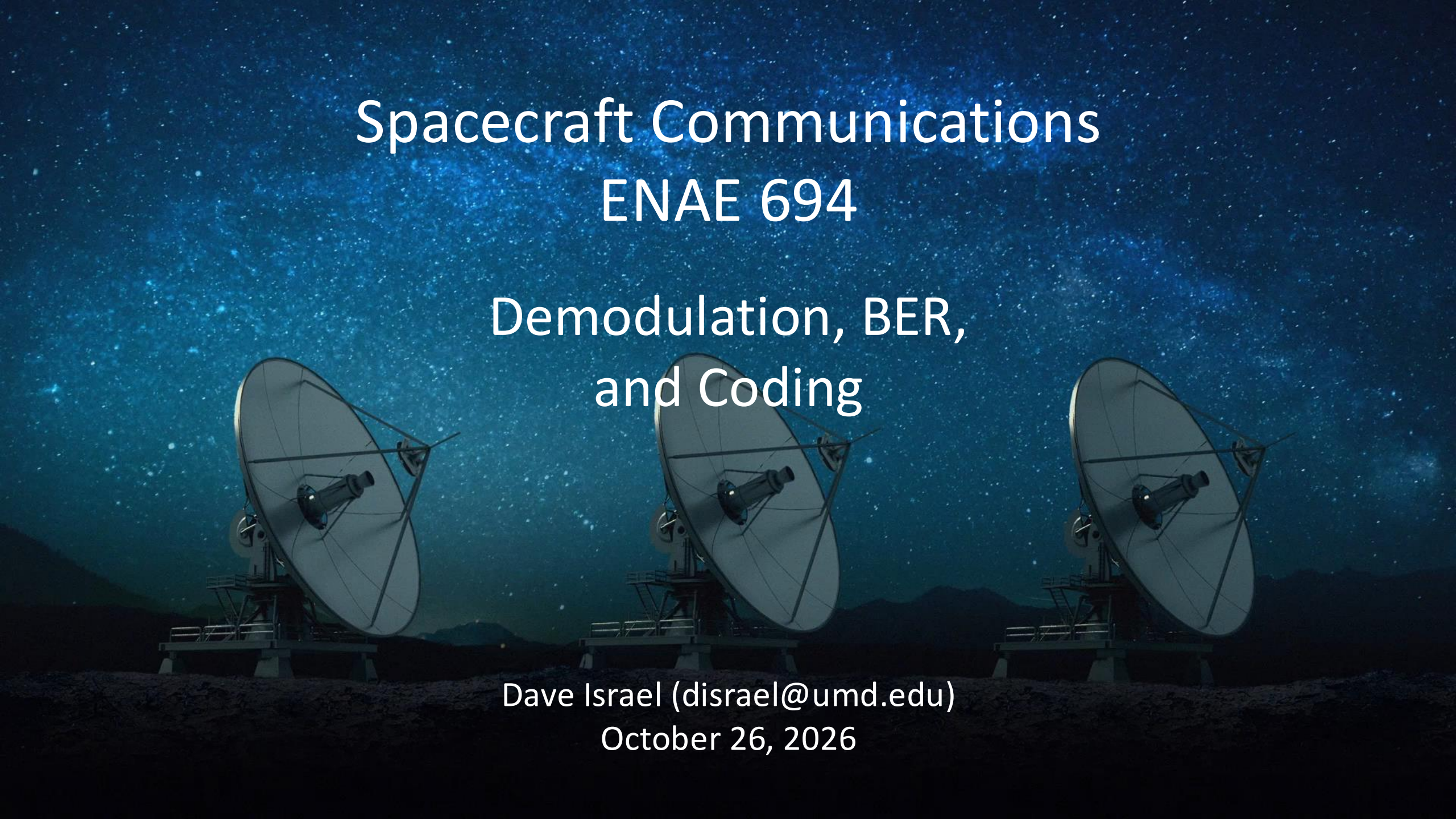


Spacecraft Communications
ENAE 694

Demodulation, BER, and Coding

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October 26, 2026

Tonight’s Engineering Trades

Recover the carrier frequency and phase, then recover symbol timing.

Choose modulation by required power and bandwidth.

Use known mathematical relationships to correct received errors.

Trade coding gain against bandwidth, complexity, and delay.

Coherent Demodulation: Recover the Reference

Infer the carrier that would be present without the information modulation.

Frequency	Track the received carrier, including Doppler and oscillator offset.
Phase	Align the reference so modulation changes can be measured.
Symbol timing	Find the boundaries and the right instant to decide each symbol.

*Timing belongs to the data symbols. Carrier recovery supplies frequency and phase. Known patterns or differential methods resolve phase ambiguity.

[3, 16, 17]

Rate Review: Bits Grouped Into Symbols

Same 12 bits in the same 2.0 µs interval: R_b = 6.0 Mbit/s

BPSK

1	0	1	1	0	0	1	0	1	0	1	1

12 symbols / 2.0 µs = 6.0 Msymbol/s

QPSK

10	11	00	10	10	11

6 symbols / 2.0 µs = 3.0 Msymbol/s

8PSK

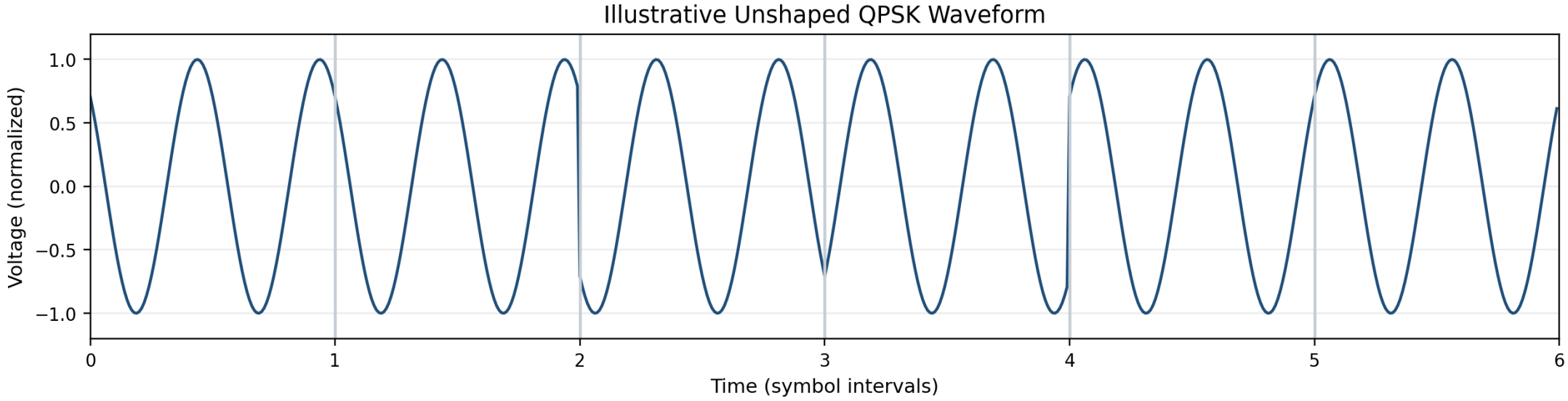
101	100	101	011

4 symbols / 2.0 µs = 2.0 Msymbol/s

$R_s = R_b / m$ $m = \log_2(M)$ bits/symbol

QPSK: Symbols, Phase, and the RF Waveform

00	00	01	11	10	10
45°	45°	135°	225°	315°	315°



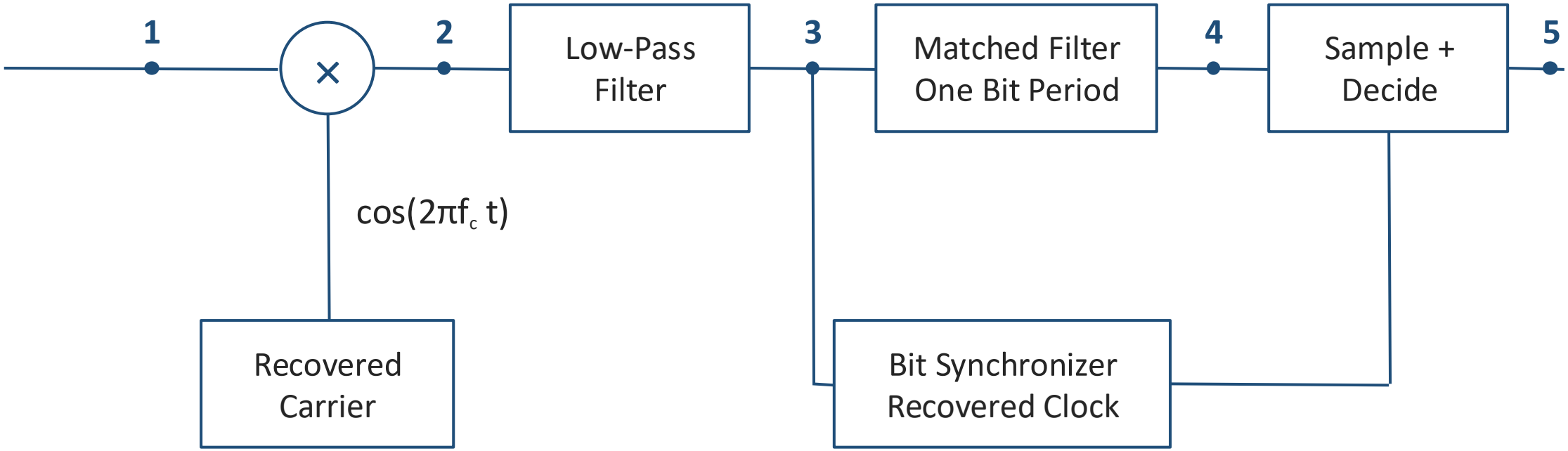
Repeated symbol: 0° change. Other boundaries: ±90° or 180°.

*Two RF cycles per symbol for illustration. Pulse shaping smooths the transitions.

[3, 6]

Receiver Walkthrough: Test Points

Received RF / IF
after the front end



Clock chooses decision instants

BPSK: 1 bit per symbol. A channel decoder follows these decisions or soft values.

Receiver Walkthrough: Why Mixing Recovers Data



m(t) = +1 or −1. Ideal phase-aligned carrier, initially without noise.

$$r(t) = A\, m(t) \cos(2\pi f_c\, t)$$

$$r(t) \cos(2\pi f_c\, t) = A\, m(t) \cos^2(2\pi f_c\, t)$$

$$= (A/2)m(t) + (A/2)m(t) \cos(4\pi f_c\, t)$$

Baseband data — Copy centered at $2f_c$

$\cos^2 x = (1 + \cos 2x)/2$. Baseband is centered at zero frequency; changing data is not constant DC.

Receiver Walkthrough: Noise Follows the Signal

1 RF input

$$r(t) = A\, m(t) \cos(2\pi f_c\, t) + n_{RF}(t)$$

2 Mixer output

$$v(t) = (A/2)m(t) + (A/2)m(t) \cos(4\pi f_c\, t) + n_{RF}(t) \cos(2\pi f_c\, t)$$

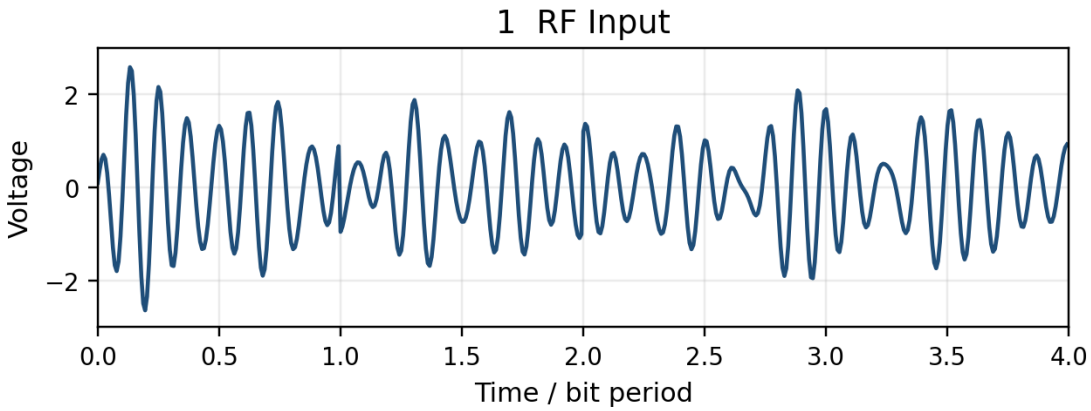
3 After low-pass filtering

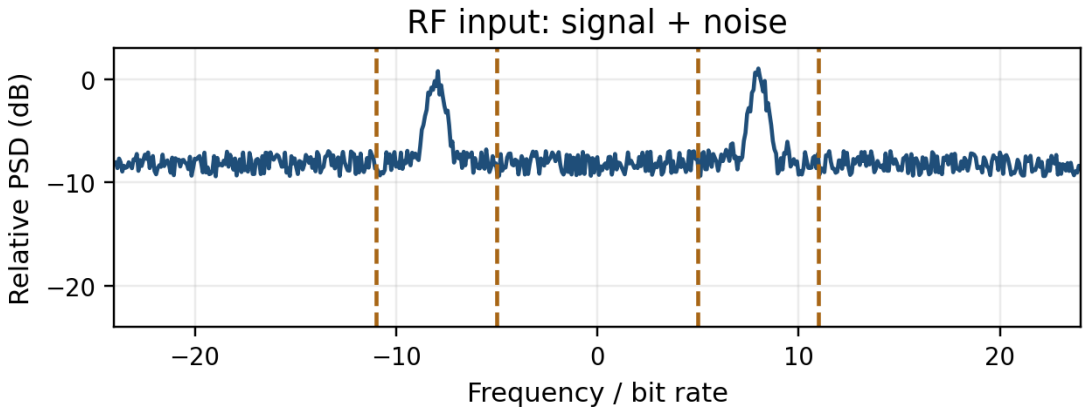
$$y(t) \approx (A/2)m(t) + n_{BB}(t)$$

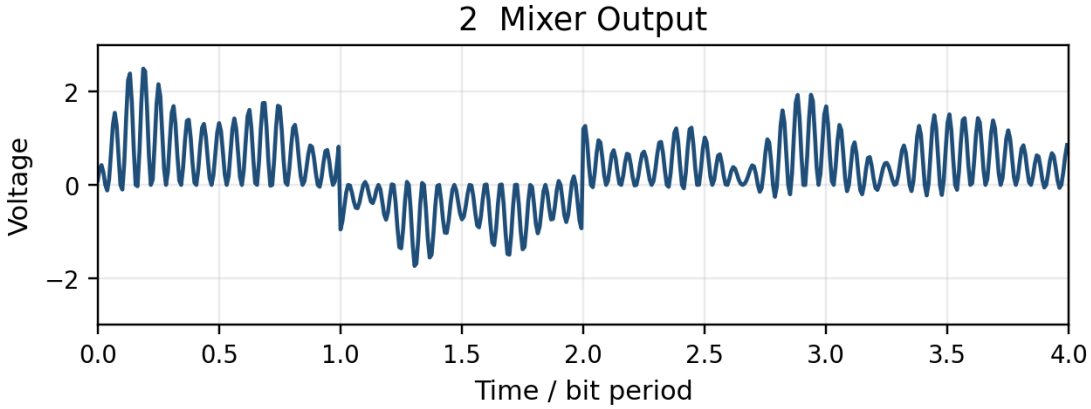
The filter rejects the $2f_c$ term and out-of-band noise. Noise translated into the data band remains. Finite receive bandwidth also rounds transitions.

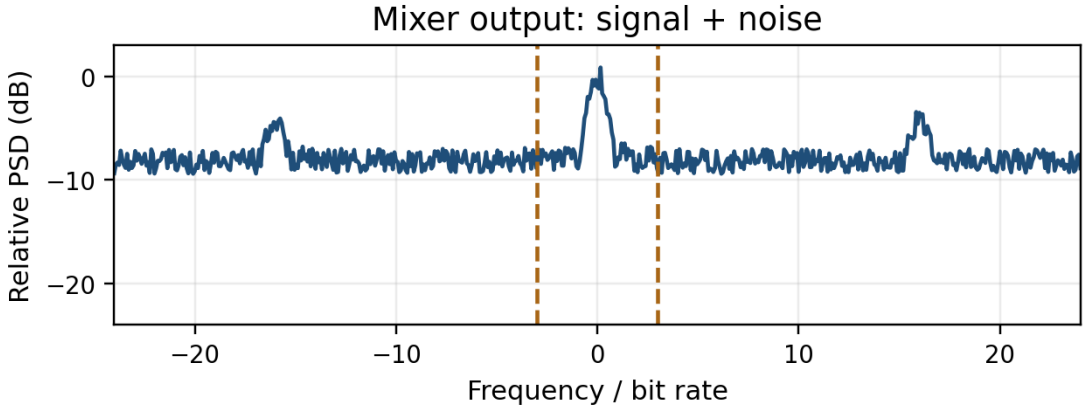
[3, 7, 14, 15]

Receiver Walkthrough: RF Input and Mixer Output





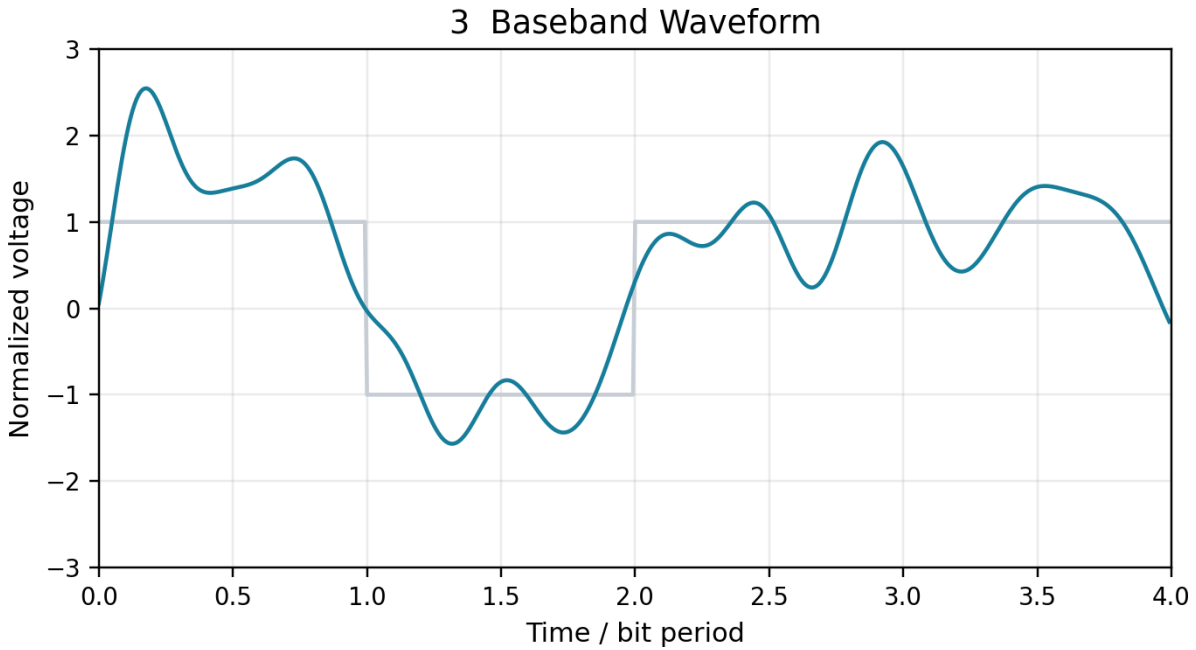


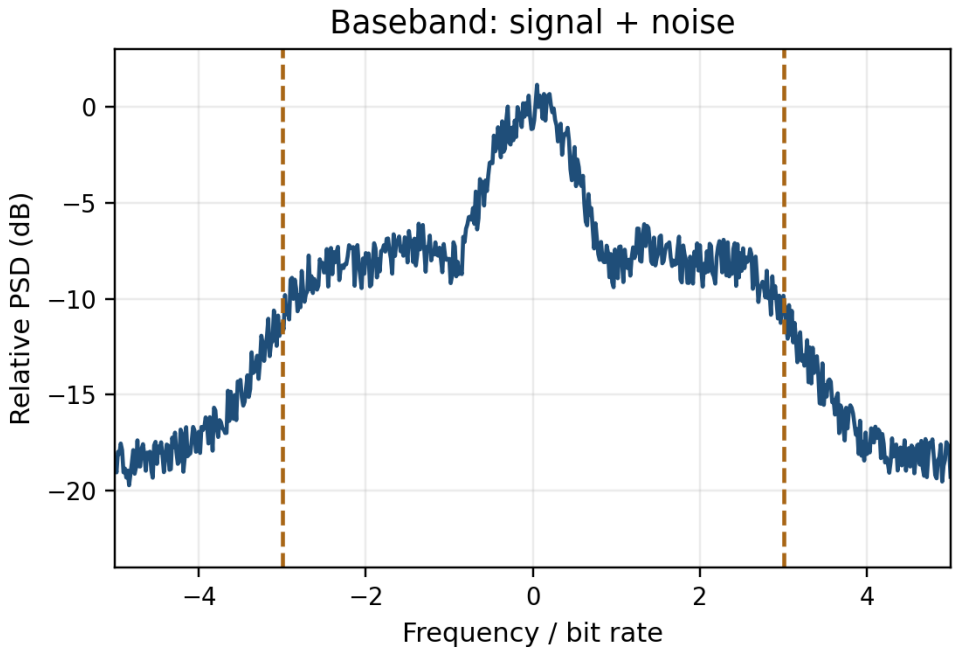


Illustrative analyzer traces, signed-frequency view. Blue includes noise; weak sidelobes are buried.
Dashed: RF passband edges at ±5 and ±11 R_β; next low-pass cutoff at ±3 R_β. carrier frequency = 8 × bit rate.

[3, 7, 14, 15]

Receiver Walkthrough: Baseband After the Low-Pass Filter





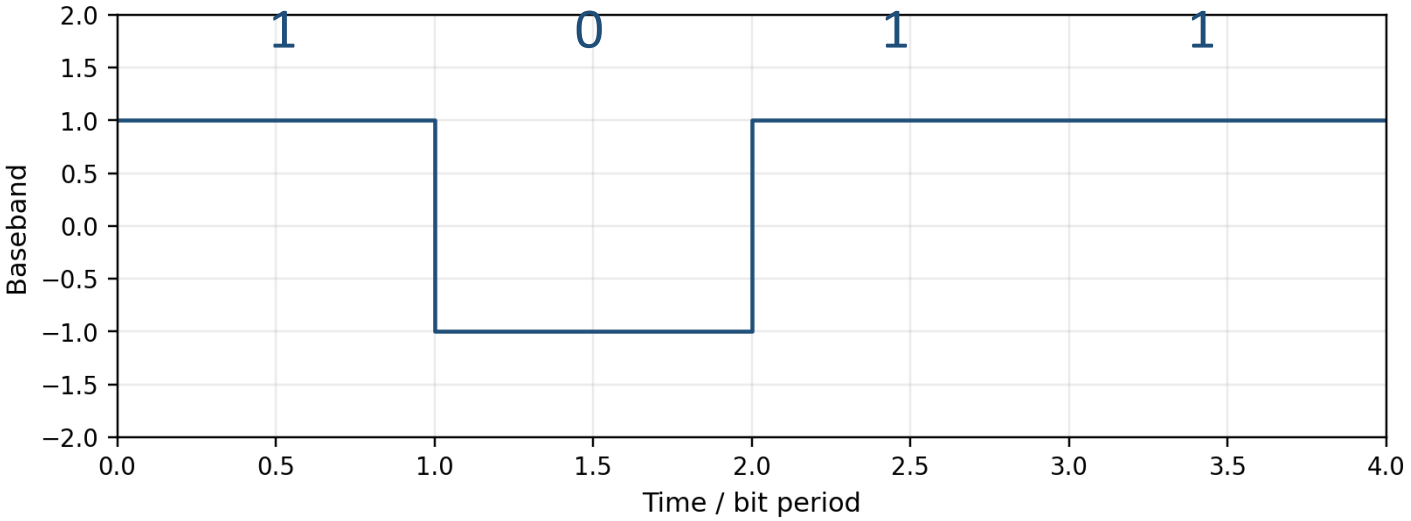
The carrier is gone. The noise in the data band is still present.

*Dashed lines: low-pass cutoff at ±3 R_β. Noise in the data band remains.
Illustrative analyzer trace includes a measurement floor; it is not a calibrated E_β/N_o measurement.

[3, 7, 14, 15]

Bit Decisions: Integrate One Bit, Then Decide

After coherent demodulation, bit 1 has positive amplitude; bit 0 has negative amplitude.



For EACH bit period:

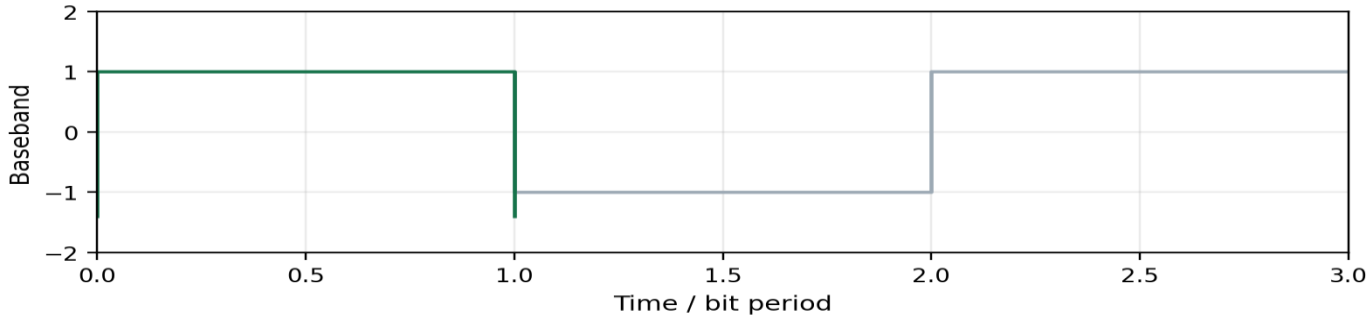
Integrate the baseband.
Decide from the sign.
Reset for the next bit.

Aligned averages:	+1	−1	+1	+1
Recovered bits:	1	0	1	1

Decision statistic = $(1/T_b)$ × integral over one bit period. Positive → 1; negative → 0.
Ideal rectangular BPSK after carrier recovery; noise omitted here to show the operation.

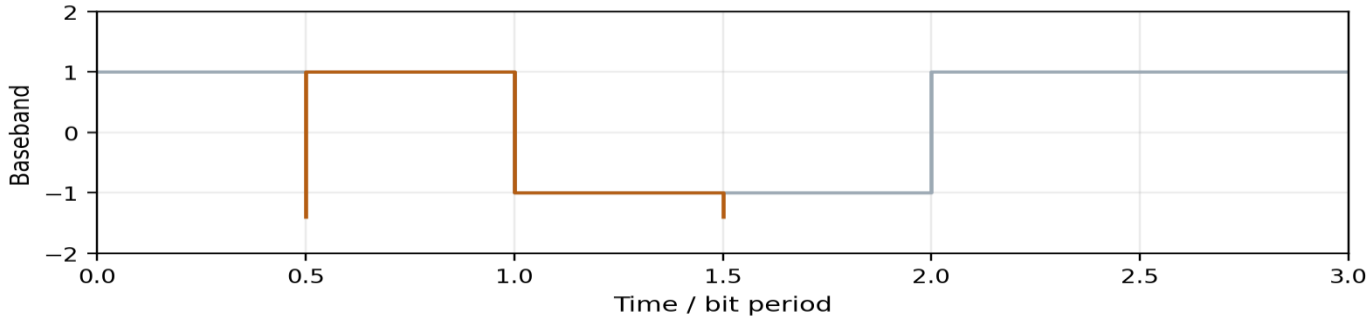
Bit Decisions: The Window Must Match the Bit

ALIGNED: integrate 0 to 1 bit period



One complete +1 bit
Average = +1
Clear decision: 1

HALF A BIT LATE: integrate 0.5 to 1.5

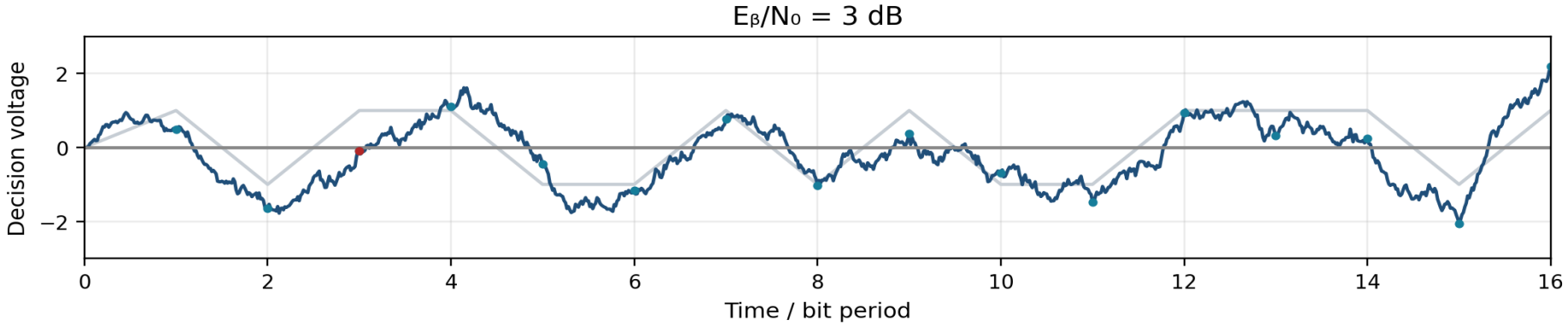


Half +1 and half −1
Average = 0
Noise can decide the bit.

Colored edges mark the integration window, not extra signal transitions. Both windows last T_b.
Bit sync aligns the window boundaries. Cancellation shown for neighboring bits of opposite sign.

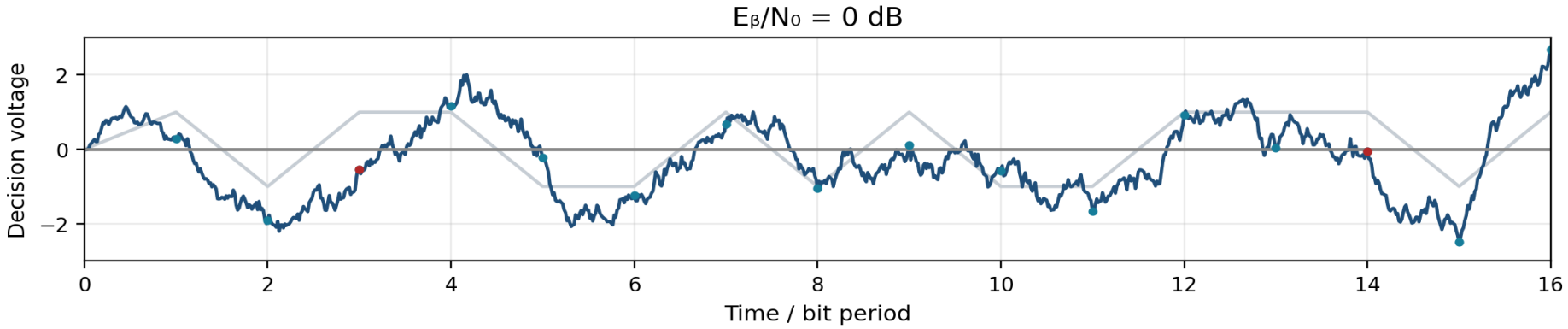
Bit Decisions: Noise Can Still Cause Errors

Sent bits: 1 0 1 1 0 0 1 0 1 0 0 1 1 1 0 1



σ = 0.50
Theory BER
2.3e-2

1 / 16 errors

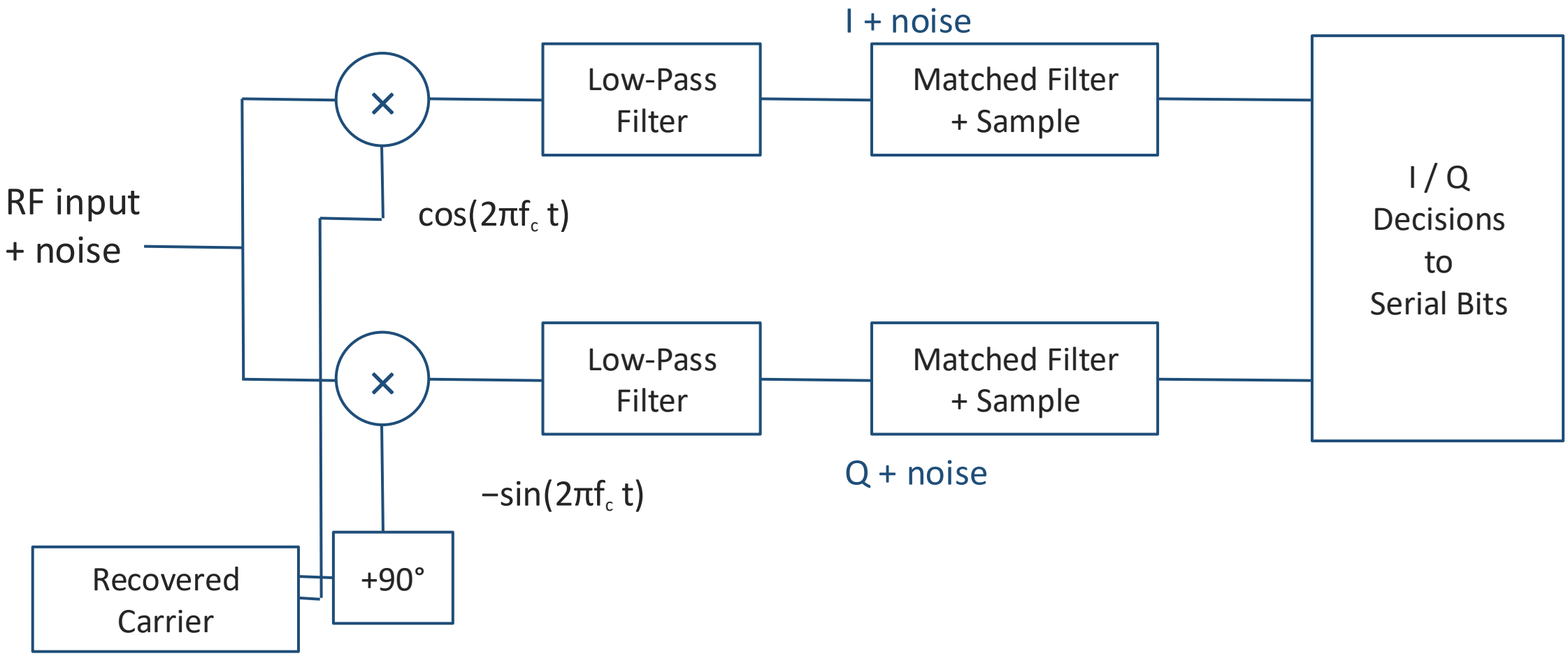


σ = 0.71
Theory BER
7.9e-2

2 / 16 errors

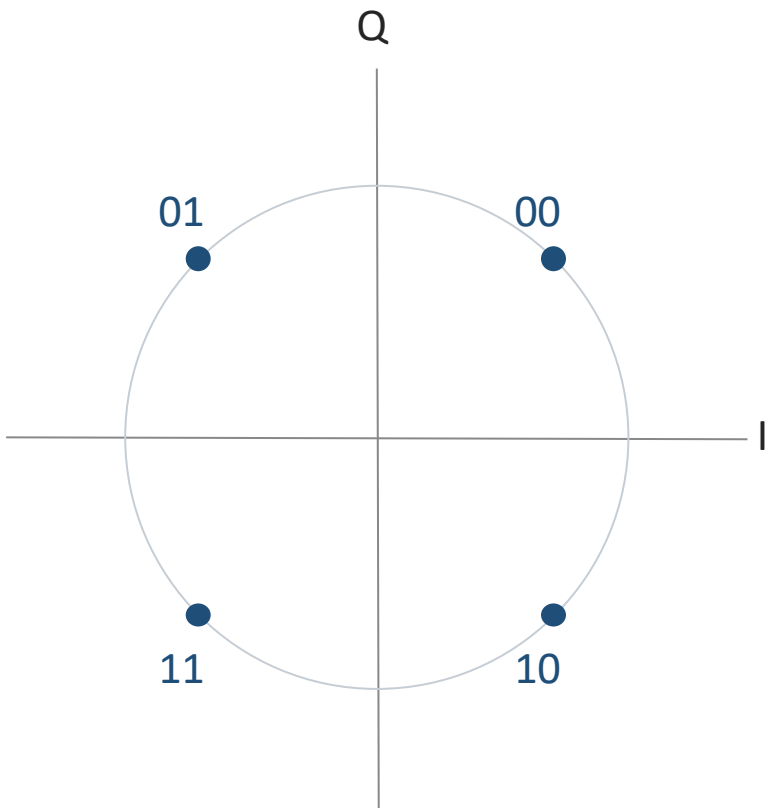
Timing aligned. Teal: sample; red: error; gray: ideal. Threshold = 0.
Noise can push the integrated value across zero and cause a wrong bit.

The I and Q View of a Demodulator



Quadrature references differ by 90°. Bit / symbol timing controls both samplers.

Constellations: Coordinates at Decision Times



Each point is one allowed decision state.

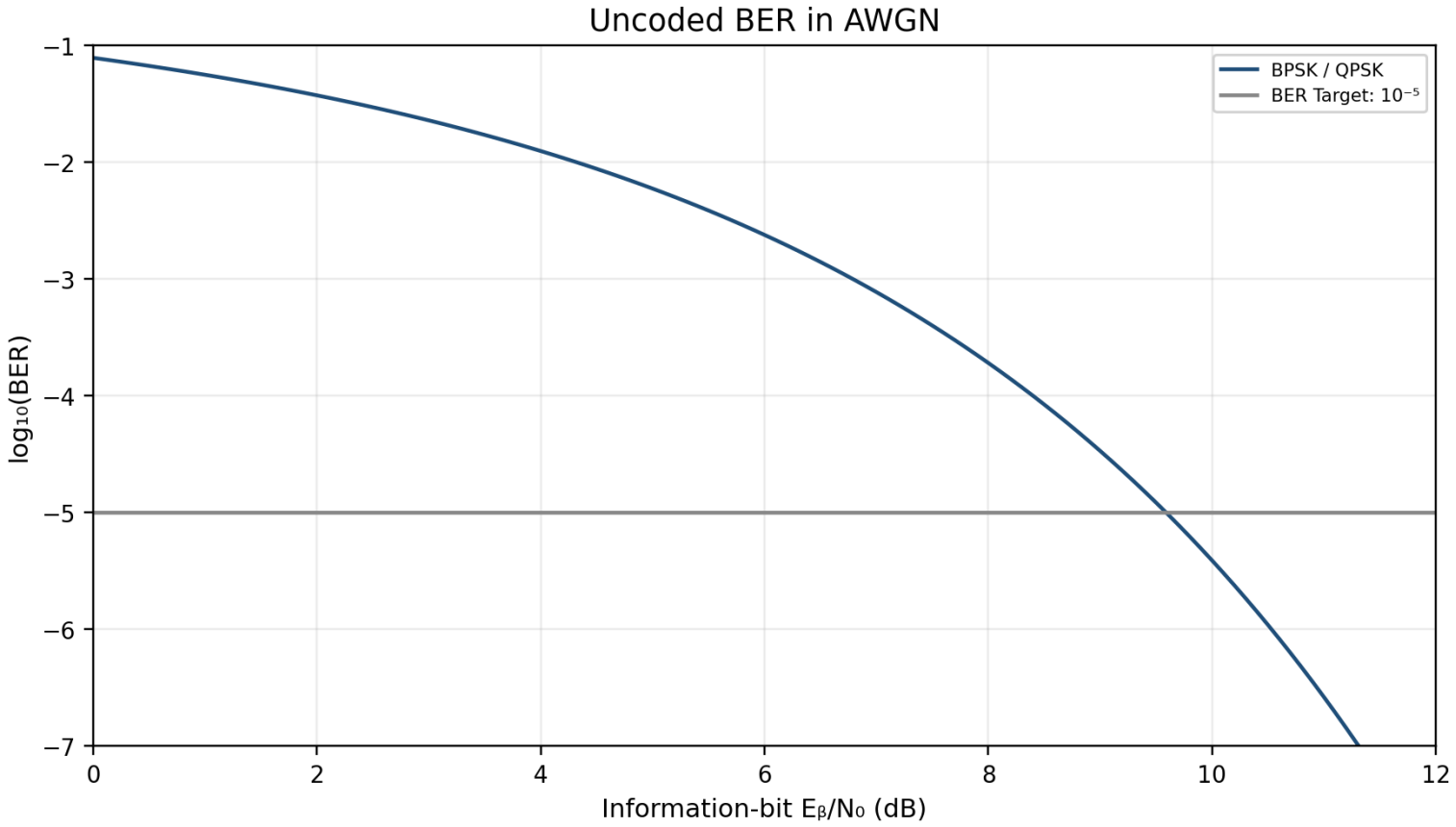
I and Q are Cartesian coordinates.

Radius represents amplitude.

Energy is proportional to radius squared.

Equal physical scale on both axes

BER Curves: Reading the Requirement



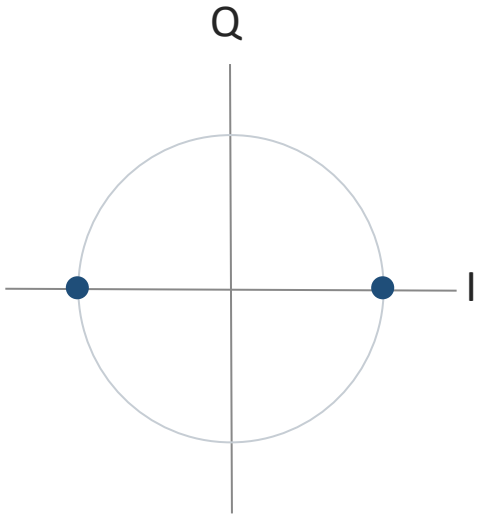
Requirement
9.6 dB

Ideal coherent detection
AWGN
Correct synchronization

−5 on the vertical axis means BER = 10^{-5}. Lower curves mean fewer bit errors.

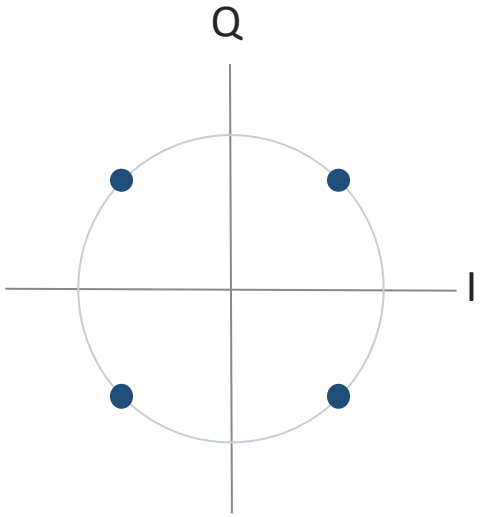
Higher-Order Modulation: More Choices

BPSK



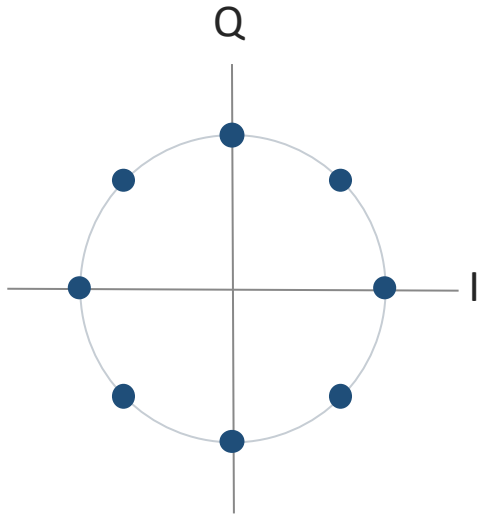
1 bits/symbol

QPSK



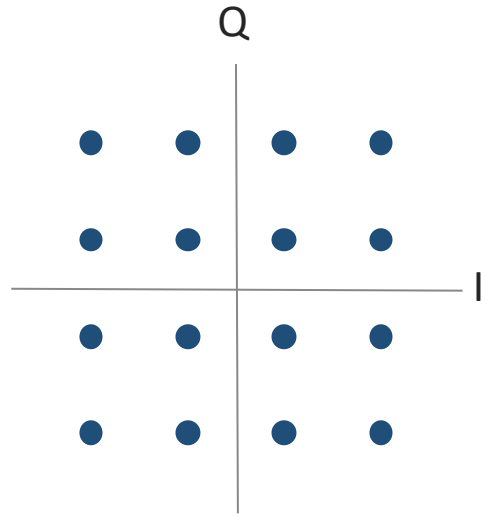
2 bits/symbol

8PSK



3 bits/symbol

16QAM

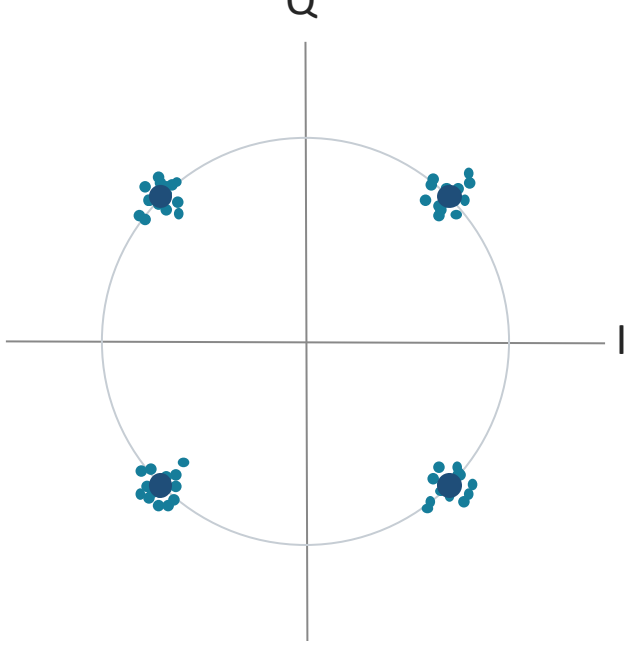


4 bits/symbol

Equal average symbol energy. More states place some decisions closer together.

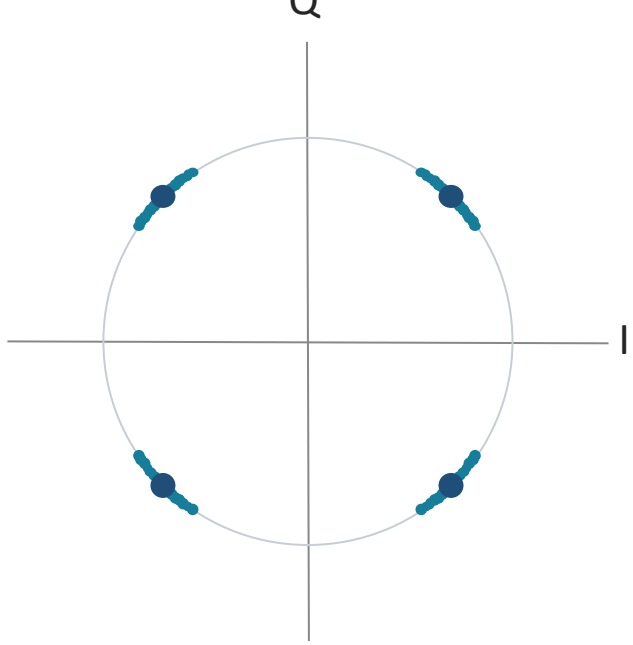
Noise and Phase Error Look Different

Additive Noise



Samples scatter around each state.

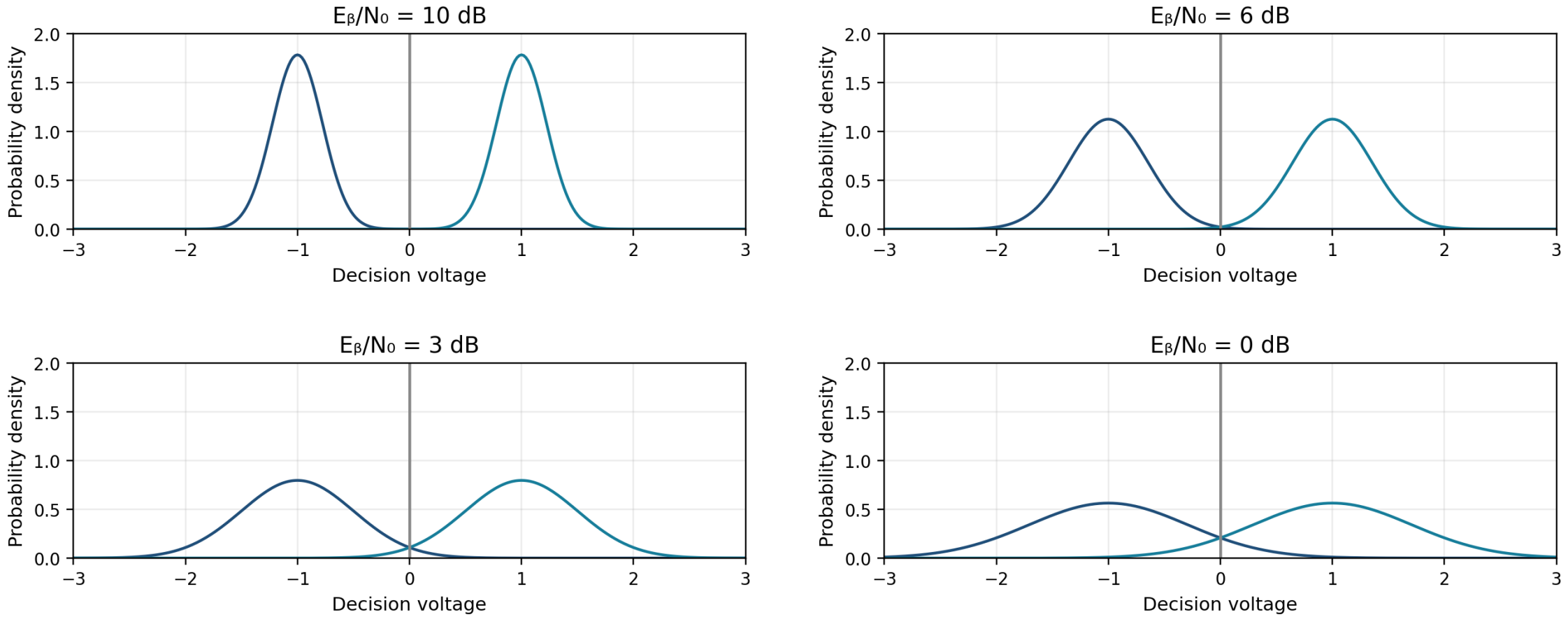
Phase Noise



Phase fluctuations spread samples around the circle.

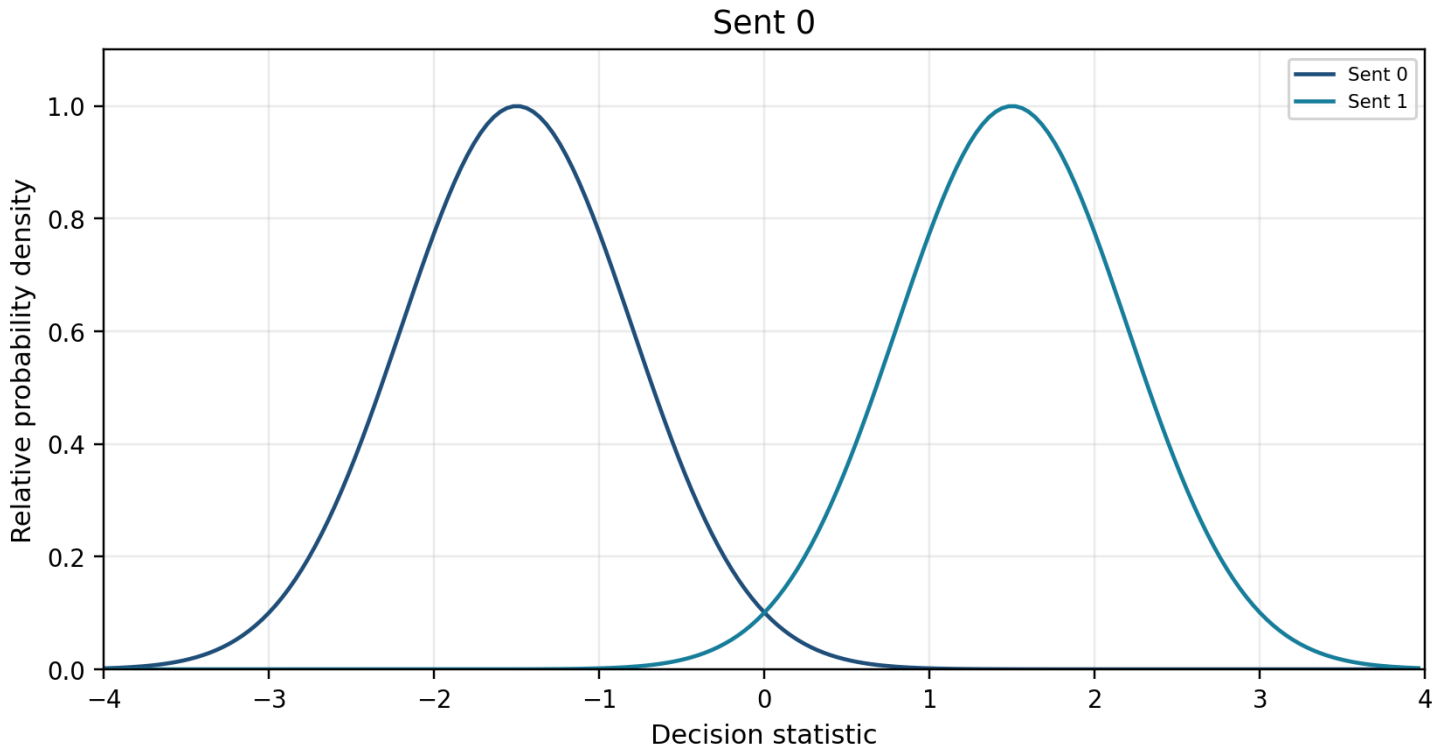
*Illustrative samples. Phase noise is an oscillator / tracking impairment, not a temperature added to T_{sys}.

Receiver Walkthrough: Overlapping Decisions



Blue: sent 0 (mean −1). Teal: sent 1 (mean +1). Wrong-side tails cause errors.

Additive Noise at the Decision



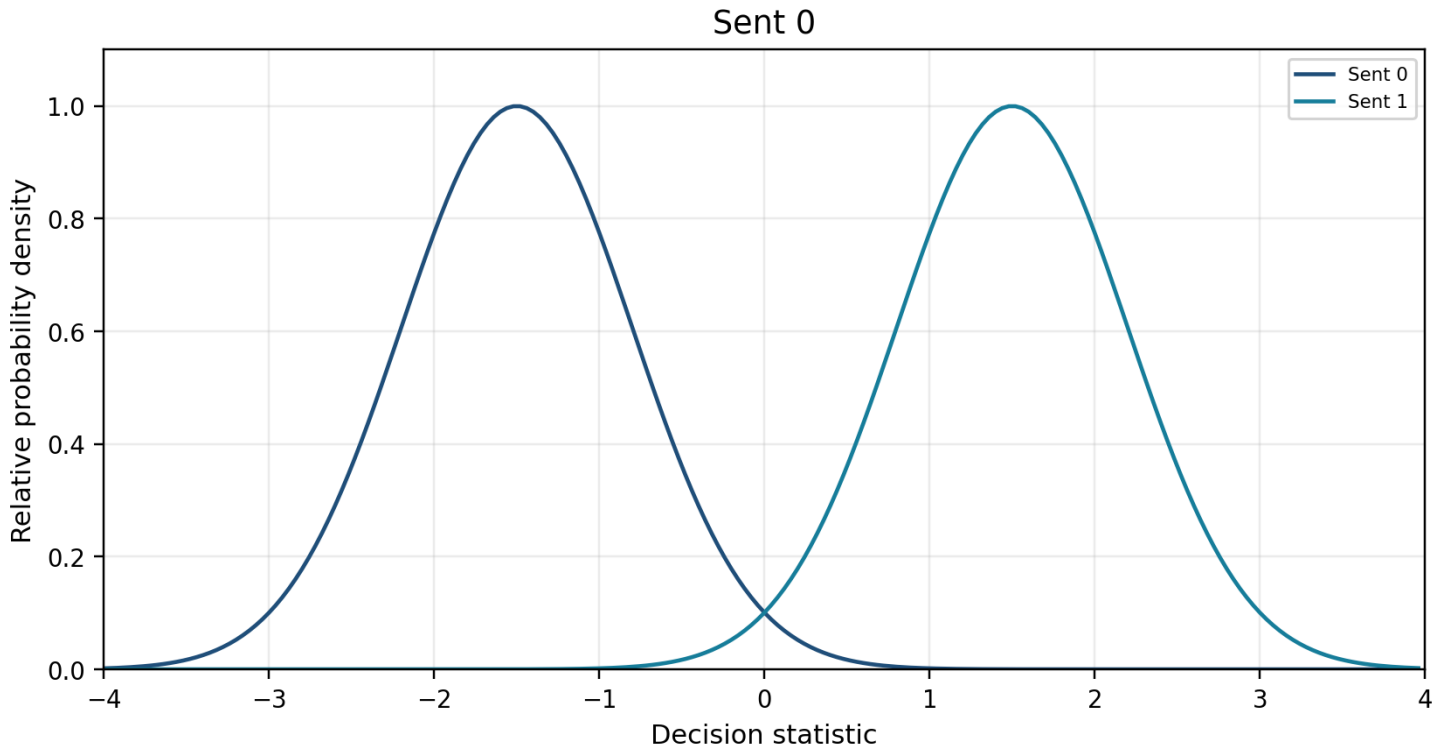
Decision threshold = 0

Noise adds a random error to each decision coordinate.

AWGN models Gaussian samples with a flat spectral density over the band of interest.

Interference and loss of lock require additional analysis.

A BPSK Error Is a Wrong-Side Decision



Decision threshold = 0

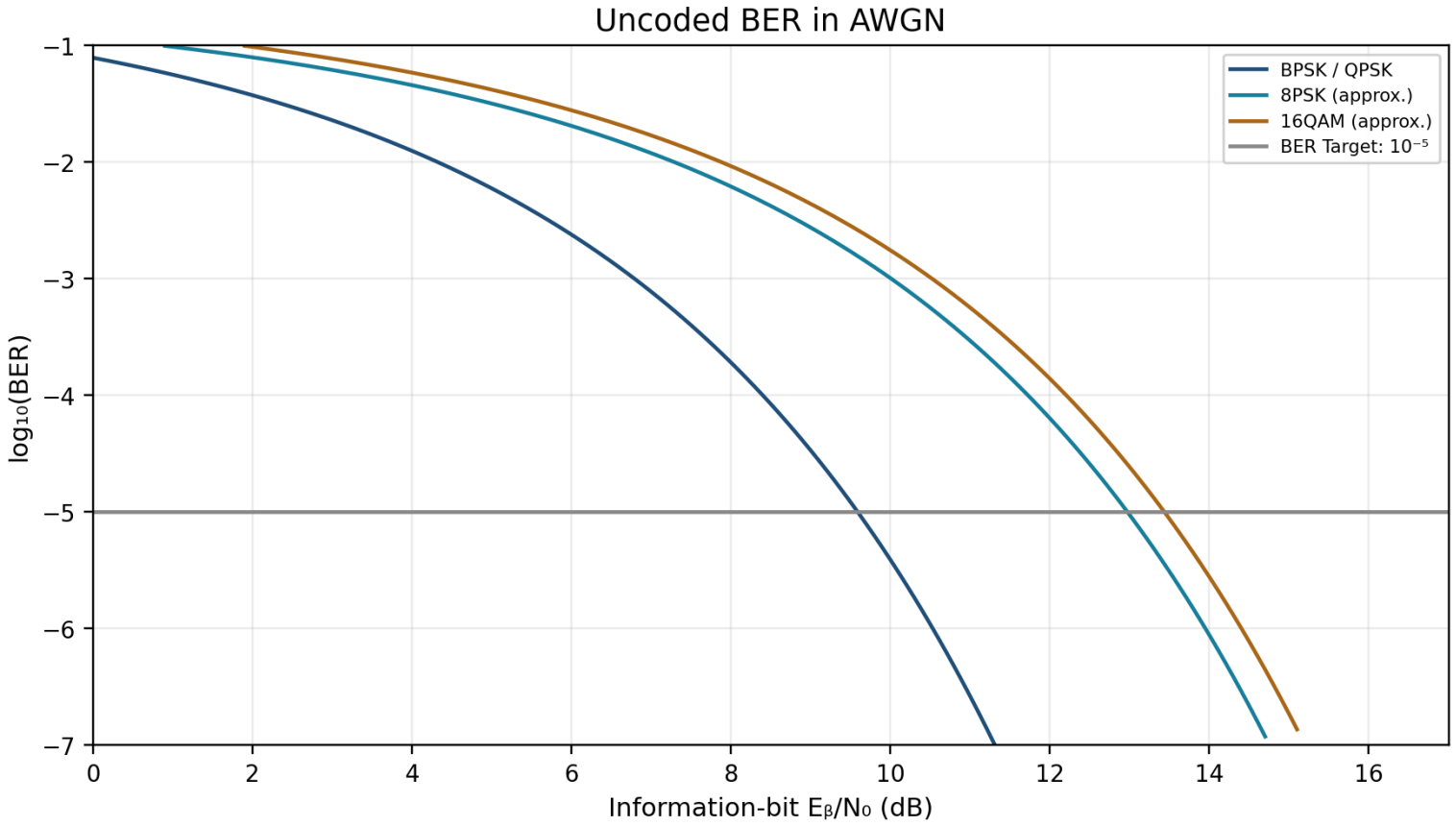
Sent 0: the decision statistic is usually negative.

Sent 1: it is usually positive.

An error occurs when noise pushes the sample across zero.

More E_b/N_0 makes wrong-side samples less likely.

BER Comparison



At BER = 10^{-5}

BPSK / QPSK 9.6 dB
8PSK 13.0 dB
16QAM 13.4 dB

Higher-order curves: Gray mapping, high-SNR approximation, AWGN

−5 on the vertical axis means BER = 10^{-5}. Lower curves mean fewer bit errors.

[7, 12]

Question: Does 8PSK Create Bit Energy?

Keep information rate, received data power, and N_0 fixed. Replace uncoded QPSK with uncoded 8PSK.

A. Available E_b/N_0 increases by 3.0 dB.

B. Available E_b/N_0 stays the same.

C. Available E_b/N_0 decreases by 3.0 dB.

Worked Answer: Does 8PSK Create Bit Energy?

B. Available E_b/N_0 Stays the Same

$E_b = S_{data} / R_b$. Neither quantity changed.

8PSK reduces symbol rate and bandwidth at this information rate.

Its closer phase states require more E_b/N_0 for the same uncoded BER.

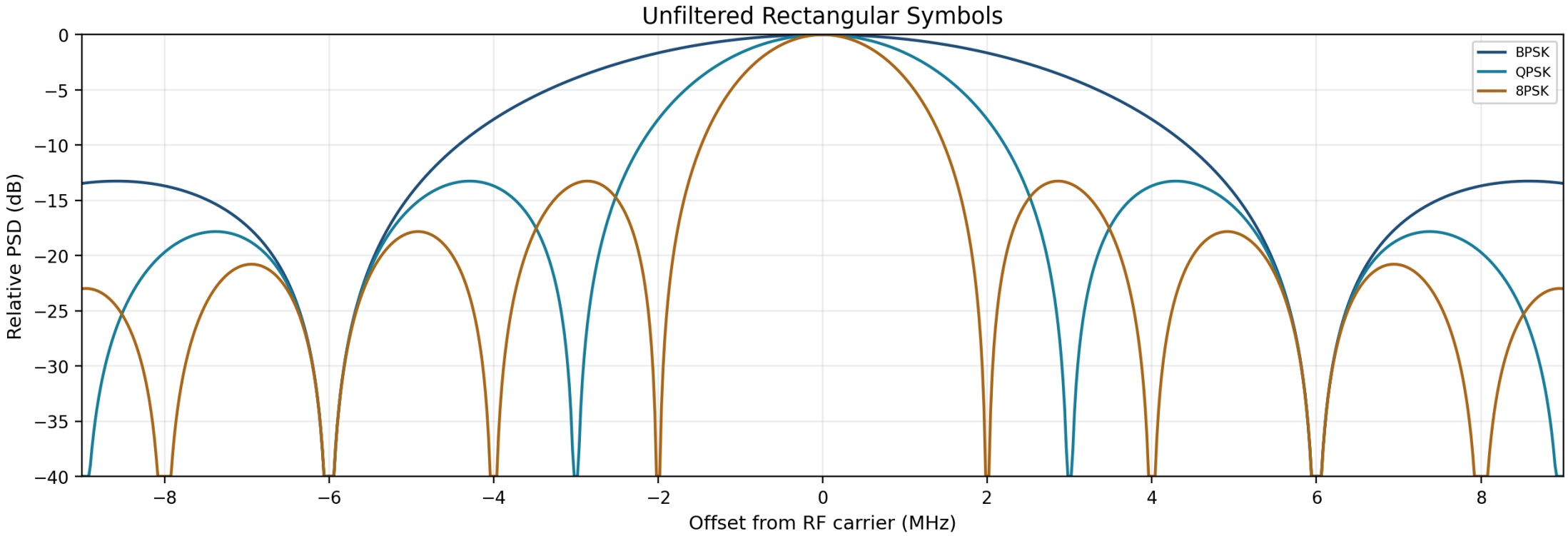
Modulation Trade: One Common Information Rate

Uncoded Format	R_s at 6.0 Mbit/s	RF Bandwidth*	Required $[E_b/N_0]$†
BPSK	6.0 Msymbol/s	12.0 MHz	9.6 dB
QPSK	3.0 Msymbol/s	6.0 MHz	9.6 dB
8PSK	2.0 Msymbol/s	4.0 MHz	13.0 dB
16QAM	1.5 Msymbol/s	3.0 MHz	13.4 dB

*Unfiltered rectangular symbols: $B_{RF} = 2R_s$, first null to first null. Sidelobes extend beyond. †BER = 10^{-5} in AWGN.

[7, 11, 12]

RF Bandwidth: The Same MHz Scale



R_b = 6.0 Mbit/s. First nulls: ±6, ±3, and ±2 MHz. $B_{RF} = 2R_s$.

*Each peak is normalized to 0 dB. Sidelobes extend beyond the stated mainlobe bandwidth.

Question: Downlink Bandwidth

Send 6.0 Mbit/s using uncoded 8PSK.

Each symbol carries 3 bits.

What is the unfiltered first-null RF bandwidth?

A. 2.0 MHz B. 4.0 MHz C. 6.0 MHz

Use $R_s = R_b / m$, then $B_{RF} = 2R_s$.

Worked Answer: Downlink Bandwidth

B. 4.0 MHz

$R_s = R_b / m = 6.0 / 3 = 2.0$ Msymbol/s

$B_{RF} = 2R_s = 2 \times 2.0 = 4.0$ MHz

8PSK fits a 4.0 MHz first-null bandwidth limit.

This step checks bandwidth. The next step checks the available bit energy.

Worked Example: Downlink Margin

For this 8PSK link: available $[E_b/N_0]$ = 17.0 dB.

Required $[E_b/N_0]$ = 13.0 dB. Implementation loss = 1.0 dB.

Margin = available − required − implementation loss

Margin = 17.0 − 13.0 − 1.0 = 3.0 dB

The link meets a 3.0 dB reserve.

At BER = 10^{-5}. Use the rounded 13.0 dB requirement supplied in the modulation comparison.

Modulation: Bandwidth Versus Required Power

At the same information rate and BER, compare both requirements.

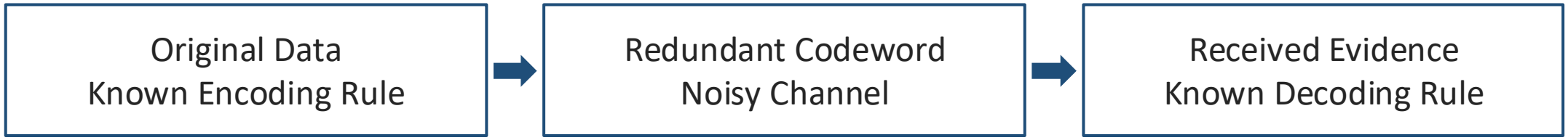
QPSK halves BPSK's mainlobe width at the same ideal E_b/N_0.

8PSK saves more bandwidth, but needs more E_b/N_0.

The useful choice fits the spectrum and the available link power.

Break

Coding: Known Math at Both Ends



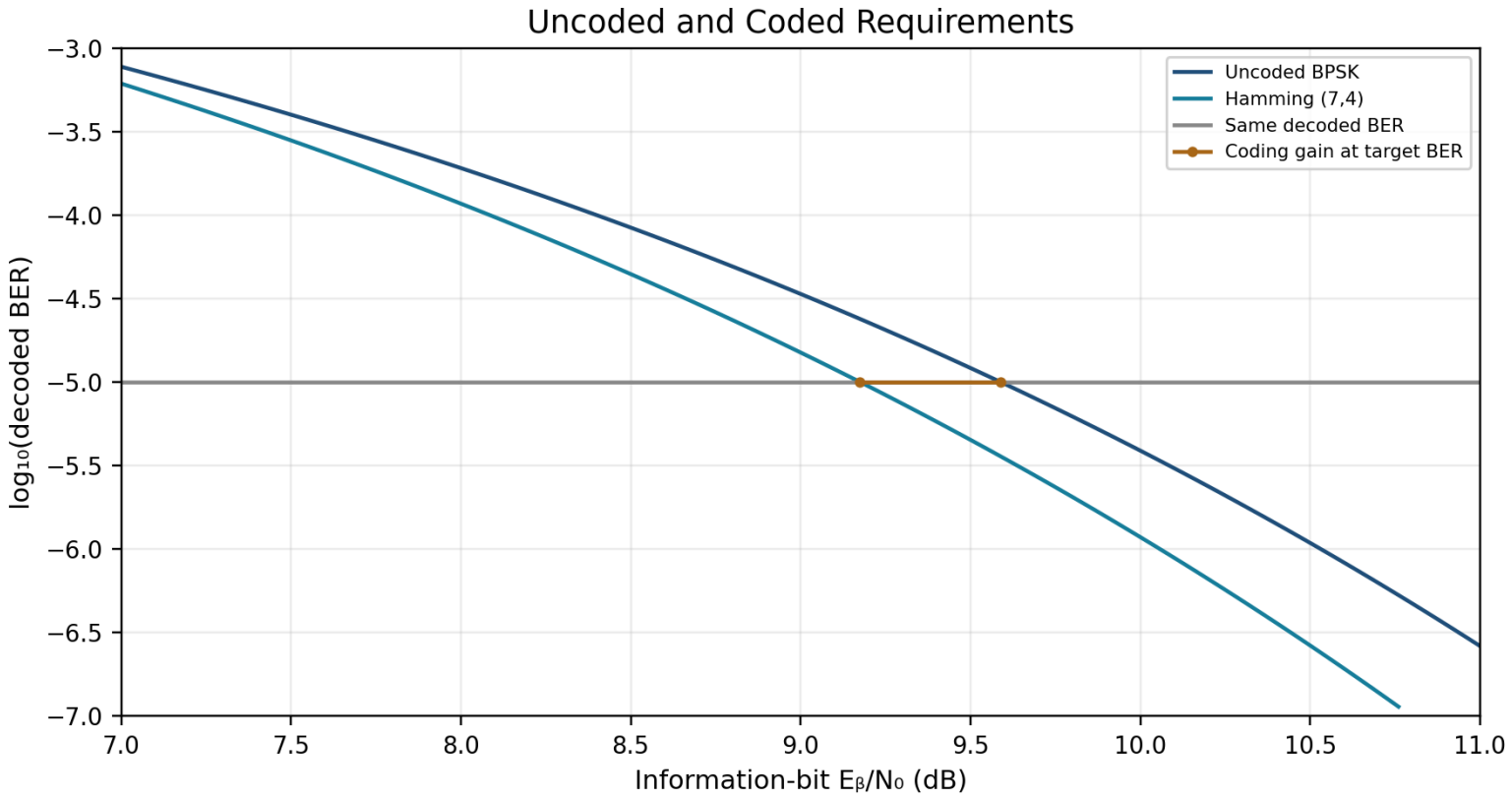
The encoder creates mathematical relationships among the transmitted bits.

The decoder tests those relationships to infer the original data.

Only selected patterns are valid. That structure makes correction possible.

*Both ends agree on the code in advance. The decoder estimates the message and can still fail when errors exceed its capability.

Coding Gain: Reading the Horizontal Difference



At BER = 10^{-5}

Uncoded: 9.6 dB
Coded: 9.2 dB

Coding gain:
9.6 − 9.2 = 0.4 dB

*Coherent BPSK, AWGN, hard-decision Hamming decoding. Exact error-pattern calculation. This small teaching code gives only a small gain.

[7, 8, 13]

Coding Gain: A Worked Link-Budget Credit

G_c = required uncoded $[E_b/N_0]$ − required coded $[E_b/N_0]$

At the same BER = 10^{-5}: G_c = 9.6 − 3.0 = 6.6 dB.

At fixed information rate, required EIRP can fall by 6.6 dB.

The receiver needs less bit energy to reach the same decoded BER.

*The 3.0 dB coded threshold is an assumed design input, not a specification for every rate-1/2 code.

Coding Gain: Power, Rate, or Margin

Use of 6.6 dB Gain*	Hold Fixed	Result
Reduce transmit power	Information rate, path, receiver	Required EIRP falls 6.6 dB
Increase information rate	EIRP, path, receiver	Up to 4.6 × rate if bandwidth permits
Increase link margin	EIRP and information rate	6.6 dB more margin

*One 6.6 dB credit to allocate. $10^{6.6/10} = 4.6$. Receiver, code, and bandwidth limits still apply.

Coded Link: What Does “Symbol” Mean?

Information: 1.0 Mbit/s enters a rate-1/2 encoder.

Coded bits: 1.0 / 0.5 = 2.0 Mbit/s leaves the encoder.

QPSK: 2 coded bits/symbol gives 1.0 Msymbol/s.

BPSK: 1 coded bit/symbol gives 2.0 Msymbol/s.

Coding documents often call each coded bit a “symbol.”
A modulation symbol can carry more than one coded bit.

Always identify which symbol is meant. Here E_b/N_0 uses information-bit energy.
Sources: CCSDS 130.1-G-3, §§3–4; CCSDS 131.0-B, convolutional coding.

Coding Reduces the Required Bit Energy

Available E_b/N_0 comes from received power and information rate.

At fixed power and information rate, coding leaves it unchanged.

A rate-1/2 code sends twice as many bits, each with half the energy.

The decoder uses the known relationships to correct errors.

The benefit: a lower required E_b/N_0 at the same decoded BER.

$E_b = S_{data}/R_b$. $E_c = rE_b$, so rate 1/2 gives $[E_c/N_0] = [E_b/N_0] - 3.0$ dB.

Question: Coding and Bit Rate

Information arrives at 4.0 Mbit/s.

A rate-1/2 encoder sends 2 coded bits for every input bit.

What coded bit rate leaves the encoder?

A. 2.0 Mbit/s B. 4.0 Mbit/s C. 8.0 Mbit/s

Code rate r = information bits / transmitted coded bits.

Worked Answer: Coding and Bit Rate

C. 8.0 Mbit/s

$R_c = R_b / r$

$R_c = 4.0 / 0.5 = 8.0$ Mbit/s

The information rate remains 4.0 Mbit/s.

For QPSK, the next step is $R_s = R_c / 2 = 4.0$ Msymbol/s.

Concatenated Rates: A Historical Example

At 1.0 Mbit/s Information	Overall Code Rate r	R_c (Mbit/s)	NRZ BPSK Mainlobe*
Uncoded	1	1.0	2.0 MHz
RS(255,223)	223/255	1.1	2.3 MHz
RS + Rate-1/2 CC	223/510	2.3	4.6 MHz
LDPC(8176,7156)	7156/8176	1.1	2.3 MHz

*First-null-to-first-null bandwidth = $2R_s$. Rectangular pulses have sidelobes outside this mainlobe.

[5, 8]

Parity Detects Errors but Cannot Locate Them

DATA	EVEN PARITY
1 1 0 0 1 1 1	1

Six 1s in the transmitted eight-bit word.

Data 1100111 contains five 1s. Even parity adds a 1.

A single flipped bit changes the parity result.

Two flipped bits can pass the parity test.

Detection may lead to discard or retransmission. FEC attempts correction.

Hamming Example: Why One Error Is Correctable

4 information bits + 3 check bits = 7 transmitted bits

Only 16 of the 128 possible seven-bit patterns are valid codewords.

Valid codewords differ in at least 3 bit positions.

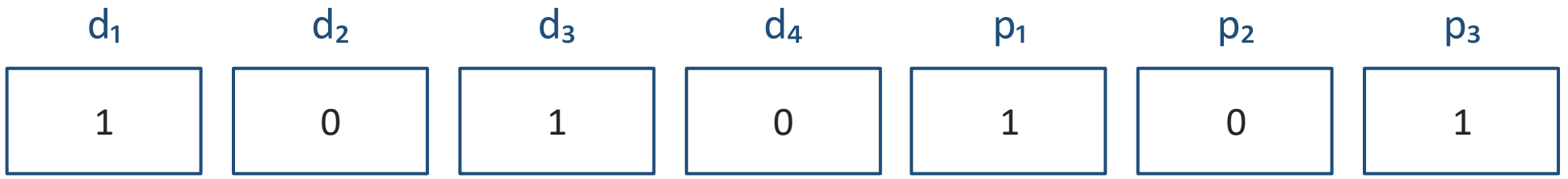
Example pair at the minimum separation:

Valid A	0 0 0 0 0 0 0	Received differs from A in 1 bit
Received	0 0 0 0 1 0 0	
Valid B	1 0 0 0 1 1 0	Received differs from B in 2 bits

One flipped bit leaves a unique nearest valid word: choose A.

Every other valid word is at least 2 bits from this received pattern.
Assumes at most one bit error. Source: Hamming (1950), Error Detecting and Error Correcting Codes.

Hamming Example: Encoding 1010



$p_1 = d_1 \oplus d_2 \oplus d_4 = 1$

$p_2 = d_1 \oplus d_3 \oplus d_4 = 0$

$p_3 = d_2 \oplus d_3 \oplus d_4 = 1$

Codeword: 1010101 $\oplus$ means XOR (addition modulo 2)

Hamming Example: Encode, Locate, Correct

1 Encode with modulo-2 arithmetic

$d = [1\ 0\ 1\ 0]$ $c = dG = [1\ 0\ 1\ 0\ 1\ 0\ 1]$

$$G = \begin{bmatrix} 1&0&0&0&1&1&0 \\ 0&1&0&0&1&0&1 \\ 0&0&1&0&0&1&1 \\ 0&0&0&1&1&1&1 \end{bmatrix} \qquad H = \begin{bmatrix} 1&1&0&1&1&0&0 \\ 1&0&1&1&0&1&0 \\ 0&1&1&1&0&0&1 \end{bmatrix}$$

2 Bit 5 flips: received $r = [1\ 0\ 1\ 0\ 0\ 0\ 1]$

3 Calculate $s = Hr^T$: each row of H selects bits to XOR

$s_1 = r_1 \oplus r_2 \oplus r_4 \oplus r_5$
$= 1 \oplus 0 \oplus 0 \oplus 0 = 1$

$s_2 = r_1 \oplus r_3 \oplus r_4 \oplus r_6$
$= 1 \oplus 1 \oplus 0 \oplus 0 = 0$

$s_3 = r_2 \oplus r_3 \oplus r_4 \oplus r_7$
$= 0 \oplus 1 \oplus 0 \oplus 1 = 0$

4 $s = [1\ 0\ 0]^T$ matches column 5 of H. Flip bit 5; recover data 1010.

XOR is addition modulo 2: $1 \oplus 1 = 0$. This example assumes one bit error.
Source: Hamming (1950), Error Detecting and Error Correcting Codes.

Hamming Example: Limits of the Promise

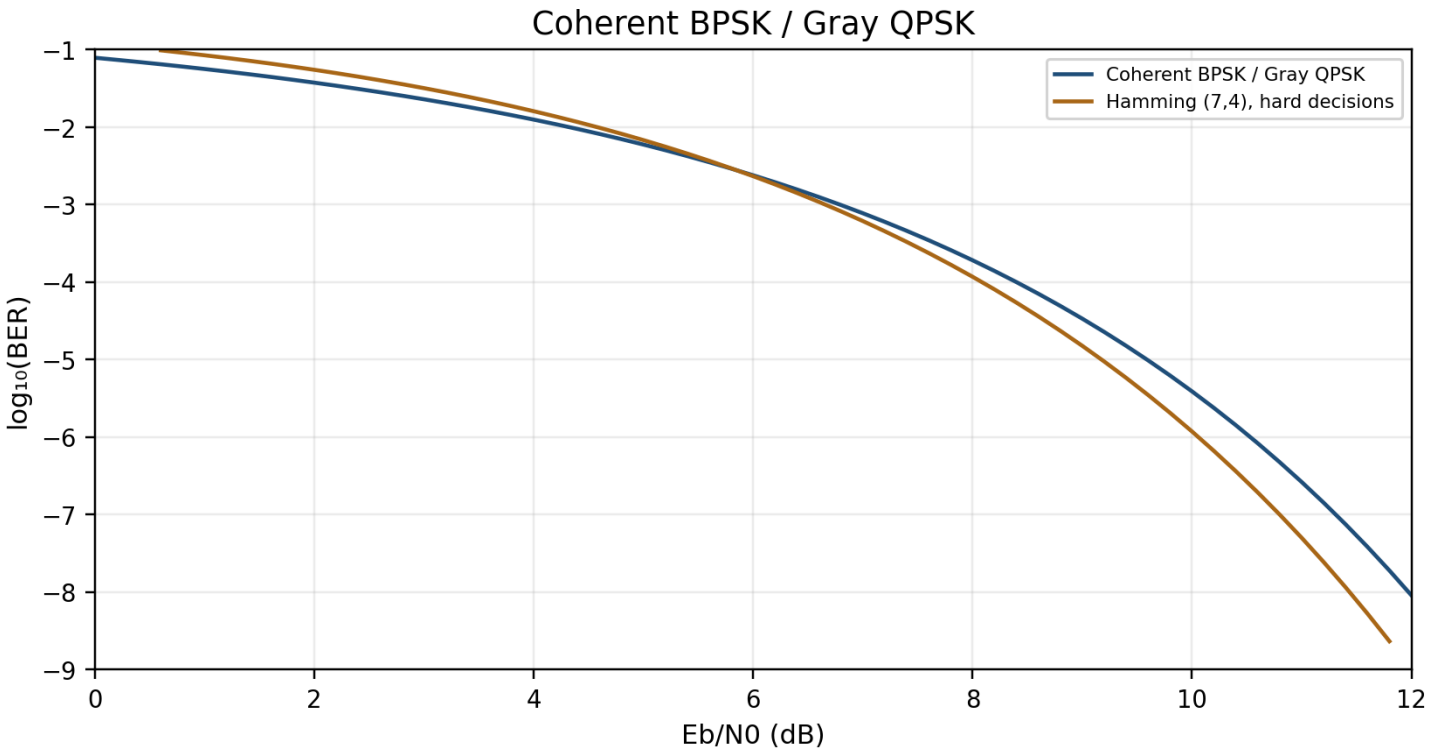
$t = \text{floor}[(d_{min} - 1)/2] = 1$ bit error correctable.

A single-error decoder can miscorrect two errors.

Zero syndrome means an allowed codeword, not proof of an error-free channel.

Coding gain must be checked at the chosen decoded BER.

Hamming (7,4): Hard-Decision BER



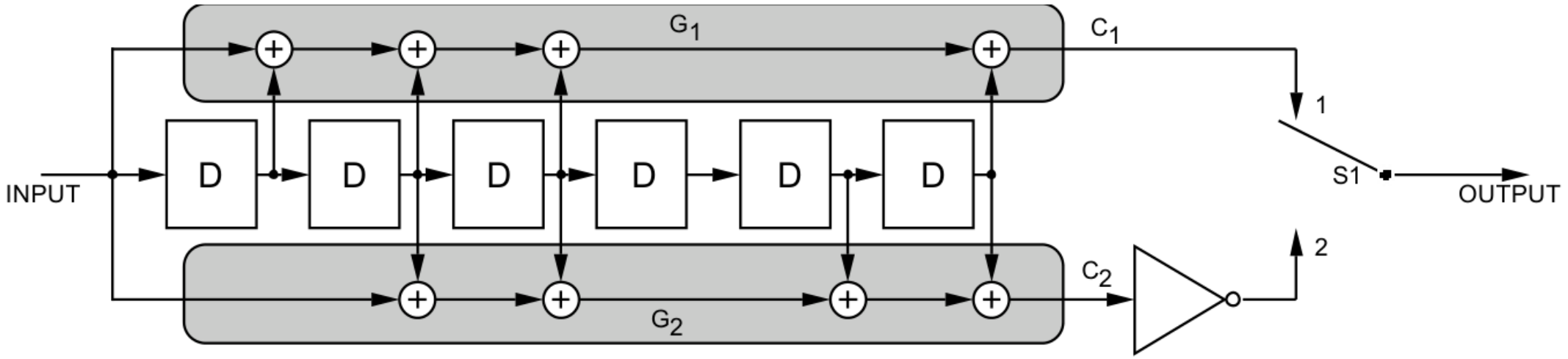
−5 on the vertical axis means BER = 10^{-5}.

Exact enumeration of all 128 possible channel error patterns.

Coherent BPSK in AWGN; r = 4/7; syndrome decoding; information-bit E_b.

A small code can have a crossover region. Coding gain depends on the chosen BER.

Convolutional Coding: Registers and XORs



D stores one bit. Each clock shifts the history one position.

$\oplus$ is XOR. The connections form two parity checks from that history.

Two output bits per input bit: r = 1/2. Current bit + 6 delays: K = 7.

S1 sends the two outputs in sequence. The lower output is inverted by the CCSDS convention.
Source: CCSDS 130.1-G-3, Fig. 4-1, p. 4-2. Viterbi decoding compares possible bit histories.

Reed–Solomon: Byte-Symbol Errors

Code	Parity Symbols	Unknown Symbol Errors Correctable
RS(255,223)	32	16
RS(255,239)	16	8

One RS symbol is one 8-bit byte, distinct from a modulation symbol. $t = (n - k)/2$ for unknown symbol errors.

[5, 8]

Reed–Solomon: Burst Length Needs Care

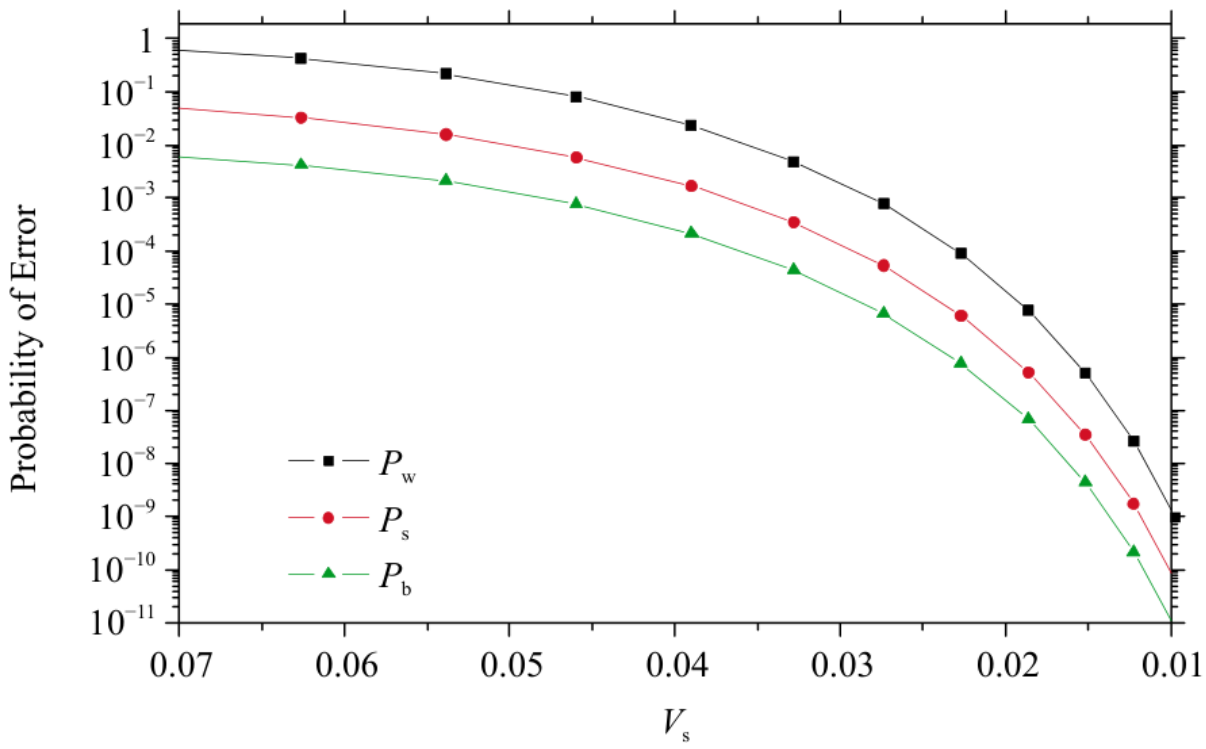
Any 16 erroneous byte symbols can be corrected by RS(255,223).

A contiguous 128-bit burst can touch 17 bytes when it is not byte-aligned.

Thus “corrects any 128-bit burst” is not a safe general claim.

Interleaving distributes a burst across multiple codewords.

Reed–Solomon: Input Errors and Output Errors



Horizontal axis: input RS-symbol error probability.
V_s decreases to the right.

Output word, symbol, and bit errors are different metrics.

This plot is not an E_b/N_0 curve.

*Source: CCSDS 130.1-G-3, Figure 5-5, p. 5-8. RS(255,223), 16-symbol correction.

Concatenated Coding: Why This Order?

TRANSMIT

RS encoder → interleaver → convolutional encoder → modulator

RECEIVE

Demodulator → Viterbi decoder → deinterleaver → RS decoder

Viterbi handles channel noise, but residual errors may arrive in bursts.

Deinterleaving spreads those bursts across different RS codewords.

RS then corrects residual byte errors: up to 16 for RS(255,223).

RS goes first at the transmitter so it acts last at the receiver. Interleaving itself corrects no errors.
Source: CCSDS 130.1-G-3, §6.

Interleaving Distributes a Burst

Word A	A1	A2	A3	A4
Word B	B1	B2	B3	B4
Word C	C1	C2	C3	C4

Transmit: A1, B1, C1, A2, B2, C2, …

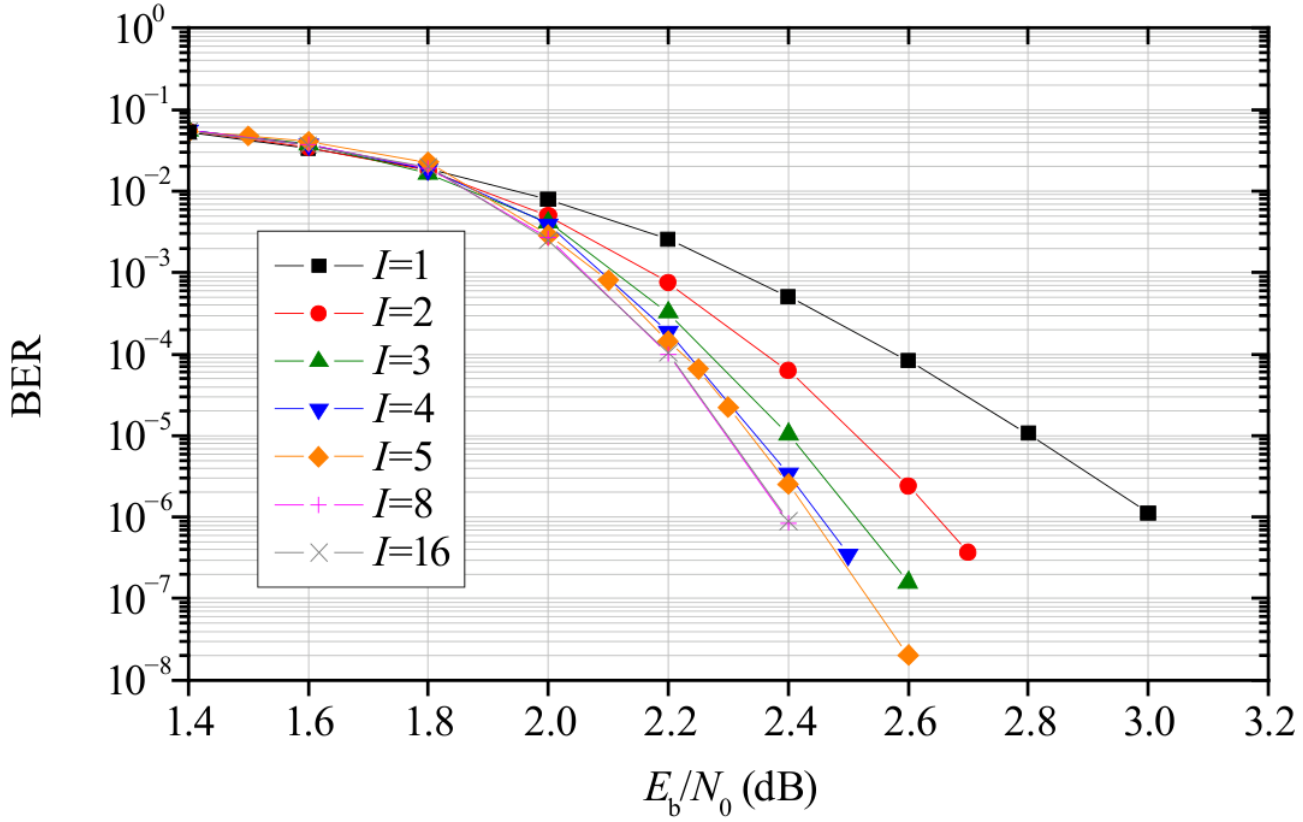
Transmit A1, B1, C1, A2, B2, C2, …

A three-symbol burst affects one symbol in each word in this example.

Interleaving spreads errors. The code supplies correction.

Memory and waiting time are costs.

Concatenated Code: Interleaving Depth



*Source: CCSDS 130.1-G-3, Figure 6-7, p. 6-9. Original simulation assumptions apply.

Outer RS(255,223), inner rate-1/2 convolutional code.

I is interleaving depth. Compare curves at the same BER.

At this operating point, deeper interleaving reduces the required bit energy.

Concatenated Coding: The Design Trade

The inner decoder handles the weak, noisy signal.

Interleaving helps the outer RS code handle remaining error bursts.

More interleaving requires more memory and adds delay.

Read coding gain at the required BER for the complete scheme.

Code rates multiply: $r_{total} = r_{RS} \times r_{CC}$. State the codes, interleaver and error metric.
Source: CCSDS 130.1-G-3, §6.

Turbo and LDPC Codes



Repeat message passing, then test parity checks.

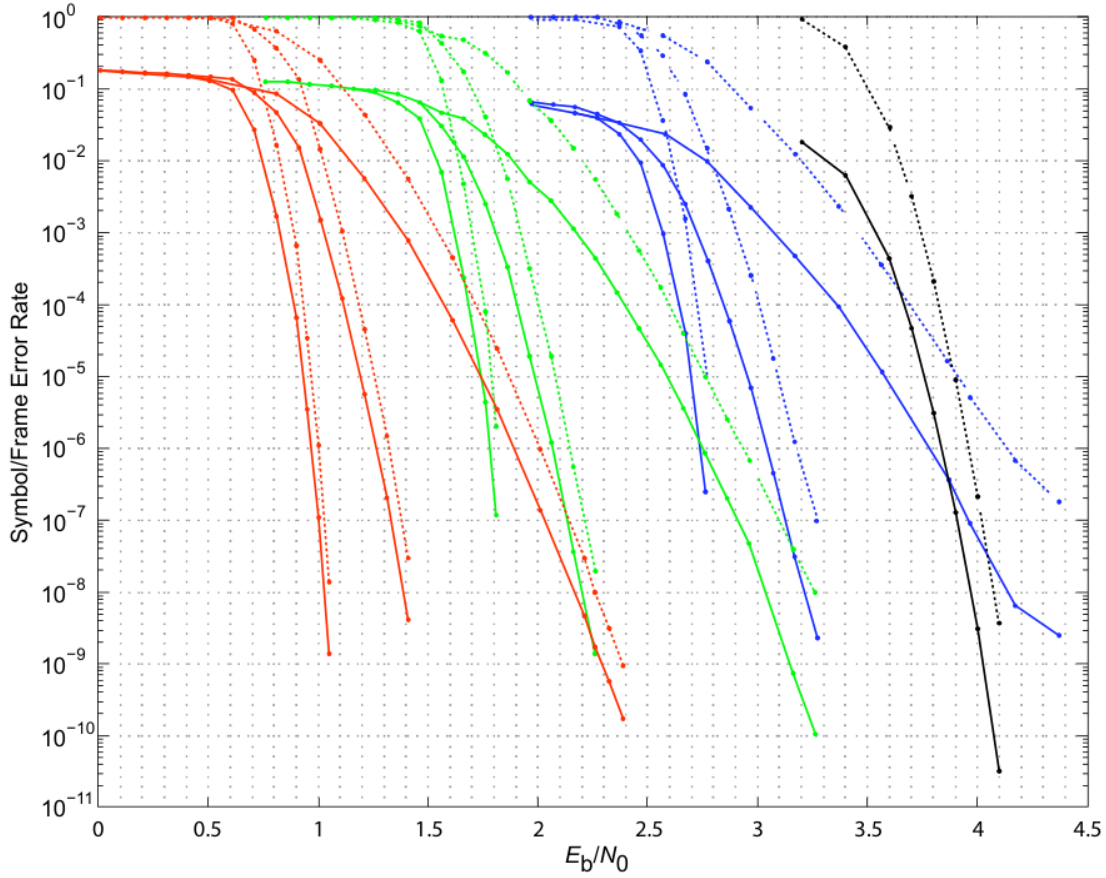
Turbo codes iteratively exchange reliability estimates.

LDPC decoders work with sparse parity checks.

Code rate, block length, iterations, and error target all affect performance.

EG-LDPC is one family within LDPC.

LDPC: Rate and Block Length Matter



Red: r = 1/2

Green: r = 2/3

Blue: r = 4/5

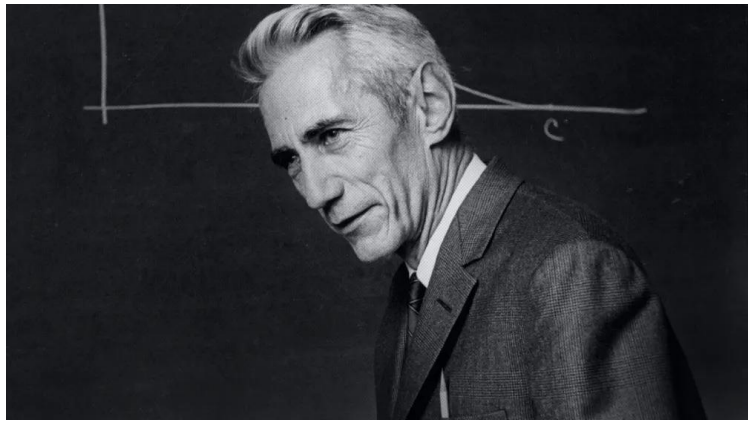
Black: C2, r ≈ 7/8

Within each color group, left to right:
k = 16,384; 4,096; 1,024
C2: k = 7,156

Solid: BER
Dashed: frame-error rate

k = information bits per block. C2 rate = 7156/8176.
Source: CCSDS 130.1-G-3, Fig. 8-8, p. 8-10. Original decoder assumptions apply.

Shannon: The Limit Behind Coding Gain



Claude Shannon, Bell Labs
1948: A Mathematical Theory of Communication

$$C = B \log_2(1 + S/N)$$

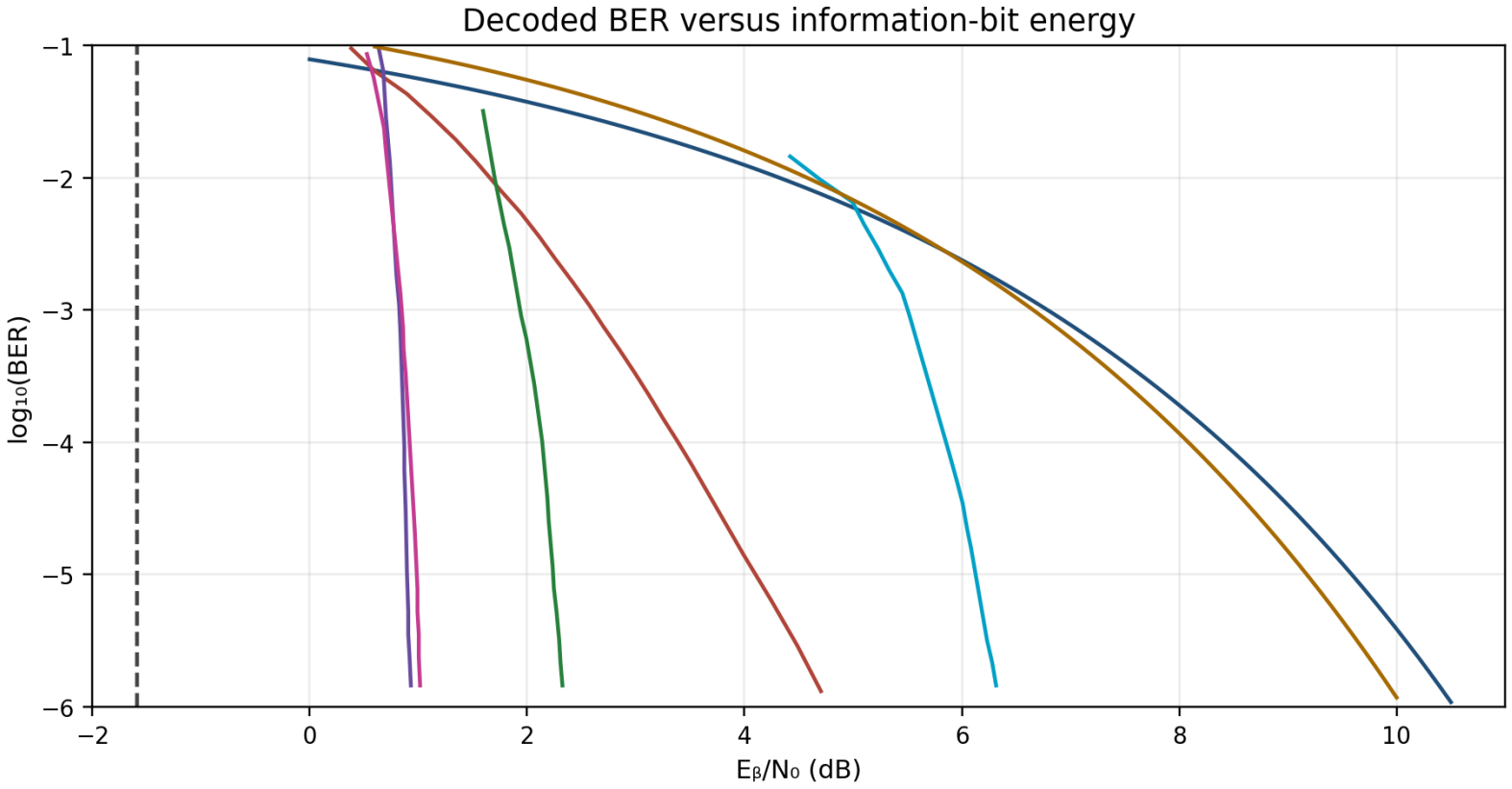
At capacity, $\eta = R_b/B$ and
$E_b/N_0 = (2^{\eta} - 1)/\eta$

As η approaches zero:
minimum $[E_b/N_0]$ approaches −1.6 dB.

A fixed modulation and code rate impose a tighter bound. Practical codes need margin.

B is the capacity-model bandwidth, not a first-null RF width.
Sources: Shannon (1948); portrait and biography: Nokia Bell Labs, Information Theory Turns 75.

BER: Coding Schemes on One Graph



Uncoded BPSK

Convolutional K=7, r=1/2

RS(255,223)

Concatenated RS + CC

Turbo r=1/2, k=8920

LDPC r=1/2, k=16384

Hamming (7,4), hard decision

Dashed: −1.6 dB ultimate wideband Shannon limit. Fixed-rate bounds are higher.

Approximate coded traces from CCSDS 130.1-G-3, Fig. 3-5, p. 3-7. BPSK/AWGN, no sync loss. Ideal interleaving for concatenation. Hamming: exact hard-decision calculation, r = 4/7.

Coding: Gain, Bandwidth, and Complexity

Benefit or Cost	What It Means for the Design
Coding gain	Less required E_b/N_0 at the same decoded error target
Bandwidth	At fixed R_b and modulation: $B_{RF} = 2R_b / (r \log_2 M)$
Complexity and delay	Decoder processing, memory, electrical power, and latency

*For this lecture: unfiltered rectangular symbols, first-null-to-first-null bandwidth. Code rate alone does not specify coding gain.

[5, 8]

Lunar Link: The Coded Candidate

Given: $[S_{data}/N_0] = 74.5$ dBHz and $R_b = 4.0$ Mbit/s.

Available $[E_b/N_0] = 74.5 - 10 \log_{10}(4.0 \times 10^6) = 8.5$ dB.

Assume a rate-1/2 decoder threshold of 3.0 dB at BER = 10^{-5}.

Margin = 8.5 − 3.0 − 1.0 = 4.5 dB ≥ 3.0 dB reserve.

The threshold excludes the separate assumed 1.0 dB implementation loss. Use information-bit energy.

Lunar Link: The Bandwidth Cost

$R_c = R_b/r = 4.0/0.5 = 8.0$ Mbit/s.

QPSK: $R_s = R_c/2 = 4.0$ Msymbol/s.

$B_{RF} = 2R_s = 2 \times 4.0 = 8.0$ MHz.

This is mainlobe width. Sidelobes extend beyond the first nulls; a spectral mask needs a separate check.

Question: Does This Link Fit?

The coded QPSK signal needs 8.0 MHz of first-null bandwidth.

The channel allows 5.0 MHz.

The power budget already passes. Can we use this design?

A. Yes B. No

Worked Answer: Does This Link Fit?

B. No. The signal is too wide.

Required bandwidth ≤ allowed bandwidth

8.0 MHz ≤ 5.0 MHz is false.

Enough received power does not remove the bandwidth limit.

Next, lower the information rate while keeping rate-1/2 coding and QPSK.

Worked Example: Maximum Rate From Bandwidth

Given B_{RF} = 5.0 MHz, r = 0.5, and QPSK (m = 2).

$B_{RF} = 2R_s = 2R_b / (r\ m)$

Rearrange: $R_{b,max} = B_{RF}\ r\ m / 2$

$R_{b,max} = 5.0 \times 0.5 \times 2 / 2 = 2.5$ Mbit/s

That is the maximum information rate for this bandwidth.

Worked Example: Maximum Rate From Power

Given $[S_{data}/N_0] = 74.5$ dBHz and required $[E_b/N_0] = 3.0$ dB.

Allow 1.0 dB implementation loss and 3.0 dB reserve.

$[R_{b,max}] = 74.5 - 3.0 - 1.0 - 3.0 = 67.5$ dB(bit/s)

$R_{b,max} = 10^{67.5/10} = 5.6$ Mbit/s

Bandwidth permits only 2.5 Mbit/s, so bandwidth controls.

Randomization, Idle Fill, and Coding

Function	Purpose	Effect on Bit Count
Randomization	Reversible XOR reduces troublesome patterns	No redundancy from XOR alone
Idle fill	Keeps a service running without user data	Consumes transmitted capacity
Channel coding	Adds relationships used for error correction	$R_c = R_b / r$

Randomization does not guarantee a transition every symbol. It is separate from encryption and FEC.

[5]

The Decisions This Lecture Enables

Demodulation: recover carrier frequency, phase, and symbol timing.

Modulation: trade required power against bandwidth.

Coding: apply known math so the receiver can infer the original data.

Code selection: trade coding gain against bandwidth and complexity.

The Complete Performance Budget

$$[E_b/N_0] = [EIRP_{data}] - [L_p] - [L_{other}] + [G/T] + 228.6 - [R_b]$$

Margin = available $[E_b/N_0]$ − required $[E_b/N_0]$ − $[L_{impl}]$

The margin must meet the required reserve.

The waveform must also fit the bandwidth and receiver capabilities.

$EIRP_{data}$ includes the data-power fraction once. [k] = −228.6 dBW/Hz/K.

Galileo: The Antenna That Would Not Open



1991: high-gain antenna failed to deploy fully.

Planned maximum: 134.4 kbps

Low-gain link at Jupiter, before improvements: 8–16 bps

What could engineers still change?

Illustration shows the intended deployed antenna, not its failed configuration. Image: NASA.
Source: JPL, Galileo Jupiter Arrival Press Kit (1995), Telecommunications Strategy.

Galileo: Each Improvement Helped the Link

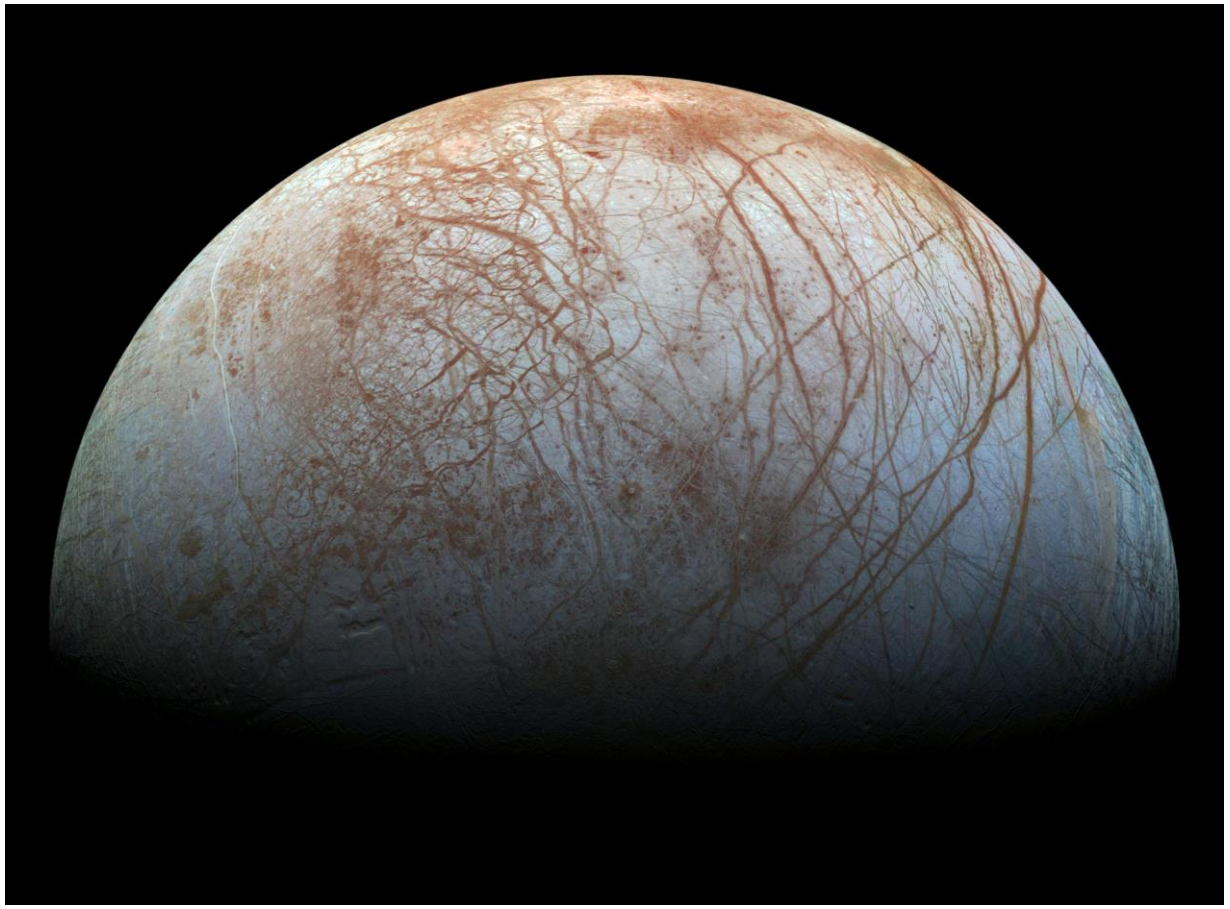
What engineers changed	Link-budget benefit	Gain*
Combined ground antennas	Higher effective G/T	Up to 4.0 dB
Canberra “ultracone” feed + LNA	Lower system noise	1.7 dB
Suppressed the residual carrier	More RF power in data	3.3 dB
Advanced coding and decoding	Lower required E_b/N_o	1.7 dB

Listed gains add to as much as 10.7 dB: about 12× power-limited rate.

Compression: fewer bits to send; no separate RF-link dB credit.

*JPL improvement-budget figures; actual benefit varied with station and mode. Coding gain is incremental.
Source: Taylor, Cheung & Seo, Galileo Telecommunications (2002), p. 21, note 4; §§4.2, 6.1.

Galileo: This Is What Came Home



Europa

A fractured, icy surface seen by Galileo's camera.

The goal was useful science returned to Earth.

Reprocessed color mosaic from Galileo SSI images. Credit: NASA/JPL-Caltech/SETI Institute.
PIA19048; some color gaps reconstructed. science.nasa.gov/photojournal/europas-stunning-surface-2/

References and Further Reading

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[4] JPL DSN 810-005, Module 206E: Telemetry General Information.

[5] CCSDS 131.0-B-5: TM Synchronization and Channel Coding.

[6] Keysight Application Note 1298: Digital Modulation in Communications Systems.

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[7] MathWorks: Analytical Expressions Used in berawgn.

[8] CCSDS 130.1-G-3: TM Synchronization and Channel Coding, Concept and Rationale.

[9] MIT 6.450, Lecture 23: channel capacity and the wideband energy limit.

[11] Analog Devices AN-922: Digital Pulse-Shaping Filter Basics.

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[12] liquid-dsp: Linear Modem Bit Error Rates, Part II.

[13] MacKay: Information Theory, Inference, and Learning Algorithms, Chapters 1 and 11.

Receiver Models and Further Reading

[14] TUM LNTwww: Matched Filter.

[15] MathWorks: AWGN Channel.

[16] MathWorks: Carrier Synchronizer. Frequency, phase, and ambiguity.

[17] MathWorks: Symbol Synchronizer. Symbol timing recovery.

References: Shannon and Galileo

Shannon (1948). A Mathematical Theory of Communication.

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